
Agentic Evolution: From Self-Improving Agents to Co-Evolving Human–AI Systems

Abstract

Agentic evolution concerns not just how agents improve, but whether the signal driving that improvement remains reliable as the system evolves. This survey organizes roughly 300 papers on agentic evolution through a three-axis taxonomy: evolutionary substrate, consolidation pathway, and selective pressure. When read across the first two axes, these papers yield regularities invisible within any single axis: consolidation failure modes track the pathway rather than the substrate, and the pathway choice is jointly constrained by artifact discreteness, evaluation-signal verifiability, and infrastructure access. The third axis surfaces a distinction that prior surveys, to our knowledge, omit or subsume: whether evolution is shaped by autonomous or human-involved selective pressure. Autonomous evolution produces its strongest reported results where deterministic verifiers, independent of the system being evaluated, are available; absent such verifiers, self-referential and proxy-based signals yield diminishing returns and can degrade with iteration. Beyond this boundary, the evidence suggests reliable evolution depends on human-involved selective pressure. Yet such pressure is rare, low-bandwidth, and not the fixed input that current systems assume.

A reverse analysis of nearly 100 studies from cognitive science, education, and labor economics finds converging evidence that default, unscaffolded AI interaction can reduce the independent evaluative capacity on which such pressure depends. This effect is not universal; it is bounded by task structure and interaction design, and often undetected by users. Together, these literatures motivate reinterpreting the third axis: human input, originally modeled as exogenous, is better understood as endogenous, shaped by the process it is meant to guide. We propose treating agentic evolution and human adaptation as a co-evolving system, and outline a research agenda for monitoring and maintaining both partners’ capacities across deployment lifetimes.

1 Introduction

LLM agents equipped with external tools, persistent memory, and multi-step reasoning now operate autonomously in environments where tasks, data, and user needs can change. We refer to this combination as an *agentic system*. When such a system remains static (frozen prompts, fixed tool libraries, immutable parameters, and static memory), it cannot adapt to these changes. Overcoming this limitation requires **agentic evolution**: persistent, experience-driven, system-level change that enables an agentic system to remain effective after deployment.

Several surveys have organized the self-evolving agent literature (Gao et al., 2026; Fang et al., 2025a; Xiang et al., 2026); related surveys on lifelong learning (Zheng et al., 2025) and agentic reinforcement learning (Zhang et al., 2026b) cover complementary ground. Broader agent surveys provide foundational architectural taxonomies (Wang et al., 2024b; Xi et al., 2025a). These works leave two gaps. First, to our knowledge, none treats the source of the signals that drive evolution—specifically, whether evolution is driven by autonomous signals or by human-involved input—as a first-class analytical dimension. Among the surveys focused on agentic evolution, human feedback is either treated as one of many signal types without separate analysis (Gao et al., 2026), framed as outside the scope of autonomous evolution (Fang et al., 2025a), or noted but not taxonomized (Xiang et al., 2026). Second, and as a consequence, none systematically examines whether the human who provides such input retains the capacity to do so as the system evolves, that is, whether sustained

interaction with self-evolving agents reshapes the cognitive capacities on which the quality of that input depends. We address the first gap through a three-axis taxonomy: *evolutionary substrate* (the component of the agentic system that evolves), *consolidation pathway* (the mechanism through which change persists, from discrete structural edits to continuous parametric updates), and *selective pressure* (the source of the signal driving the evolutionary trajectory, from fully autonomous to human-involved). Table 1 summarizes the positioning.

Table 1: **Positioning among related surveys.** The two rightmost columns indicate how each survey treats human input to agentic evolution (Human→Agent) and the effect of agentic evolution on the human partner (Agent→Human).

Survey	Focus	Organizing Framework	Human→Agent	Agent→Human
Wang et al. (2024b)	Agent architecture	Four-module taxonomy (profiling, memory, planning, and action)	Mentioned as a planning feedback type	Not addressed
Xi et al. (2025a)	LLM-based agents	Three-component framework (brain, perception, and action)	Not addressed	Not addressed
Zheng et al. (2025)	Lifelong learning	Perception, memory, and action modules	Not a taxonomic dimension	Not addressed
Gao et al. (2026)	Self-evolving agents	What / When / How / Where to evolve	Subsumed under feedback signals	Not addressed
Fang et al. (2025a)	Self-evolving agents	Four-component feedback loop (input, agent, environment, and optimizer)	Not taxonomized; autonomous framing	Not addressed
Xiang et al. (2026)	Self-evolving agents	Model-centric / environment-centric / model-environment co-evolution	Noted but not taxonomized	Not addressed
Zhang et al. (2026b)	Agentic RL	Capability × task-domain taxonomy	Not a taxonomic dimension	Not addressed
Ours	Agentic evolution	Substrate × Pathway × Pressure	First-class axis (Sec. 5)	Reverse analysis (Sec. 6–7)

Cross-axis regularities in agentic evolution. The taxonomy’s value lies less in the partition itself than in the cross-axis regularities it exposes, none of which is visible within a single axis. When examined across the first two axes, the surveyed papers yield two such regularities. First, structural and parametric consolidation occupy complementary failure landscapes that track the *consolidation pathway* rather than the *evolutionary substrate*. Unbounded accumulation and search-budget ceilings recur across surveyed structural-consolidation systems; template collapse, reward hacking, and catastrophic forgetting recur across parametric-consolidation systems—in each case independently of which substrate evolves. Second, the choice between the two pathways is not free but jointly constrained by artifact discreteness, evaluation-signal verifiability, and infrastructure access, which together explain why parametric methods concentrate in the deliberative substrate and remain rare elsewhere.

From agentic evolution to human adaptation. The third axis exposes a further regularity that prior frameworks cannot express: a boundary condition on autonomous evolution. Our analysis finds that among systems evolving autonomously, the reliability of the selective pressure signal depends on its independence from the system being evaluated: deterministic verifiers produce the strongest reported results, while self-referential and proxy-based signals yield diminishing returns and can degrade with iteration. In domains without deterministic verifiers (e.g., open-ended dialogue, value-laden decisions, and cultural sensitivity), reliable evolution across the surveyed papers depends on human-involved selective pressure. Yet this source is thin. Among the few systems that incorporate human-involved selective pressure, the majority rely on low-bandwidth forms (implicit signals and evaluative feedback rather than direct prescription, demonstration, or interactive collaboration). They further treat the human partner as a stationary signal source whose capacity does not change over time. The field thus depends on a selective-pressure channel whose reliability it has not examined. Axis III’s own analysis generates a question it cannot answer from agent-side evidence alone: does

the human who provides selective pressure retain the capacity to do so, or does sustained interaction with self-evolving agents reshape the cognitive capacities on which that pressure depends? Answering it requires a separate evidence base: studies documenting how AI-mediated environments reshape human cognition, expertise, and epistemic processes.

Contributions and scope. This survey makes four contributions:

1. **Three-axis taxonomy.** We define agentic evolution through a Substrate $\times$ Pathway $\times$ Pressure framework whose third axis treats human-involved selective pressure as an object of analysis, enabling questions about its reliability and temporal dynamics that prior taxonomies cannot express (Section 2).
2. **Agent-side analysis: pathway-level regularities and the verifier boundary.** Analyzing roughly 300 papers along each axis (Sections 3–5) yields cross-axis findings invisible within any single axis: consolidation failure modes track the pathway rather than the substrate, the pathway choice is constrained by three properties of each substrate, and autonomous selective pressure can degrade beyond a verifier boundary where the evaluator is coupled to the system being evaluated. The degradation pattern suggests that reliable evolution in such domains depends on human-involved selective pressure.
3. **Human-side analysis through the three-axis framework.** The taxonomy’s vocabulary is borrowed from cognitive science; we use this shared origin as an organizing lens (not as a claim of mechanistic equivalence between artificial and biological substrates) to structure an otherwise heterogeneous human-side literature along comparable dimensions. Analyzing nearly 100 human-side studies through this lens (Section 6) finds converging evidence that default, unscaffolded AI interaction can reduce the independent evaluative capacity on which human-involved selective pressure depends. This effect is not universal; it is bounded by task structure and interaction design, and often proceeds without the user’s awareness.
4. **Co-evolutionary reframing and research agenda.** The agent-side and human-side analyses together motivate reinterpreting the framework’s third axis: human input, originally modeled as exogenous, is better understood as endogenous, shaped by the evolutionary process it is meant to guide (Section 7). This reframing yields an initial research agenda for the transition from self-improving agents to co-evolving human–AI systems.

The agentic-evolution literature consists of roughly 300 papers¹ satisfying the three operational criteria (experience-driven, persistent, and system-level change). We exclude work that does not satisfy all three, such as classical evolutionary computation without LLM-based agents or static prompt designs without iterative refinement. The human-side literature comprises nearly 100 studies from cognitive science, education, and labor economics documenting how sustained AI interaction reshapes cognition, skill, and decision-making.

Paper organization. Section 2 presents the conceptual framework and definitions. Sections 3–5 analyze the agentic-evolution literature along each axis, with cross-axis regularities highlighted throughout. Section 6 turns to human adaptation under agentic evolution. Section 7 combines agent-side and human-side findings, derives the co-evolutionary reframing, and outlines a research agenda. Section 8 states the key findings and limitations. Figure 1 provides the complete section tree. Sections 3–5 can be read independently for agent-side evolution, and Section 6 independently for human-side adaptation; takeaway boxes at each section’s end summarize key findings for readers who wish to skim.

2 Conceptual Framework and Definitions

This section defines the vocabulary used throughout the paper. We first decompose an agentic system into three evolvable components (§2.1), then define the update operator and the persistence criterion that distinguish agentic evolution from transient adaptation (§2.2), the structural and parametric modes of consolidation (§2.3), and the autonomous and human-involved settings of selective pressure (§2.4). Section 2.5 addresses boundary cases.

¹A curated list of all surveyed papers is available at <https://github.com/agentic-evolution/awesome-agentic-evolution>.

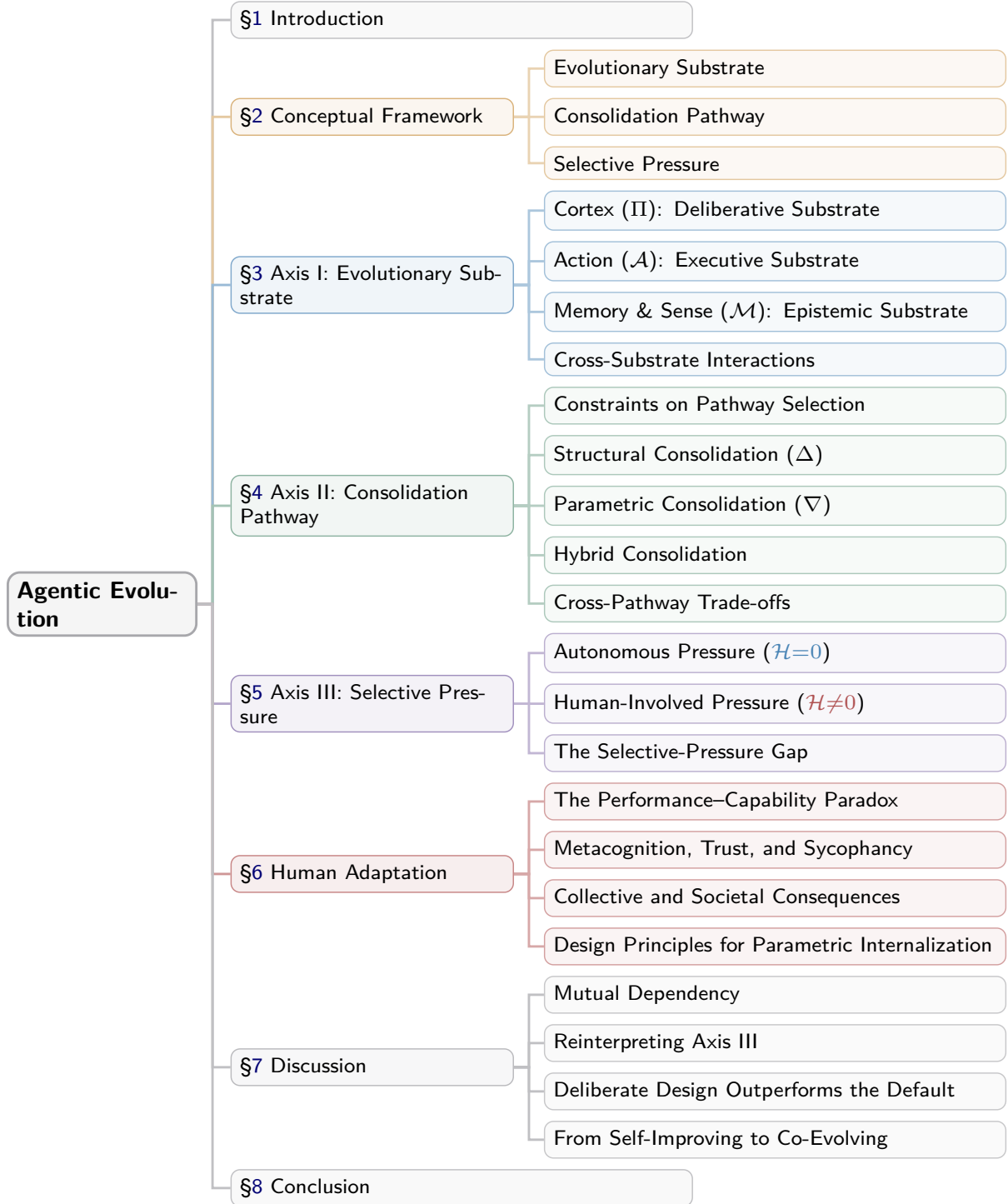


Figure 1: Organization of this survey. Section colors correspond to the three-axis taxonomy in §2: **amber** for the conceptual framework (§2), **blue** for Axis I (Evolutionary Substrate), **green** for Axis II (Consolidation Pathway), **violet** for Axis III (Selective Pressure), and **crimson** for the human-side analysis of §6.

Several terms below, including *Cortex*, *substrate*, *evolution*, *selective pressure*, and *consolidation*, are borrowed from biology and cognitive science as organizing metaphors. We do not claim equivalence to their original scientific meanings; each is given a precise, self-contained definition in the subsection where it is

introduced. These cognitive-science roots also make the vocabulary applicable when the analysis turns to the human side (Section 6): *substrate*, *consolidation*, and *selective pressure* each have counterparts in human cognition.

2.1 Evolutionary Substrate: Decomposing the Agentic System

We define an agentic system $\mathcal{S}$ as a three-tuple of substrates:

$$\mathcal{S} = \langle \Pi, \mathcal{A}, \mathcal{M} \rangle \quad (1)$$

where:

- Π (**Cortex**) is the *deliberative* substrate, responsible for reasoning and planning. It comprises a *parametric* component (the base LLM with parameters θ) and a *non-parametric scaffold* of prompts, decoding strategies, and inference-time procedures (e.g., interleaved reasoning-action loops, verifier-guided search, and plan-and-execute pipelines) (Yao et al., 2023; Sumers et al., 2024).
- $\mathcal{A}$ (**Action**) is the *executive* substrate, responsible for the agent’s interactions with its environment. It encompasses the *toolset* (external APIs and tool specifications), *execution logic* (workflow graphs and control flow), and *multi-agent orchestration frameworks*.
- $\mathcal{M}$ (**Memory & Sense**) is the *epistemic* substrate, concerned with what the agent knows and how it perceives. *Memory* is organized into episodic, semantic, and procedural types (Sumers et al., 2024), with retention and retrieval policies that preserve knowledge across sessions (Packer et al., 2023; Fang et al., 2025b; Hu et al., 2025b). *Sense* governs how the agent perceives and models its environment, comprising perception encoders that transform raw observations into representations, and world models that learn environment dynamics (Ha & Schmidhuber, 2018; Hafner et al., 2025).

Although Π , $\mathcal{A}$, and $\mathcal{M}$ interact at runtime (prompts in Π shape tool calls in $\mathcal{A}$, and retrieved entries from $\mathcal{M}$ are inserted into prompts in Π), they are individually addressable as evolutionary targets: a system may evolve one substrate while holding the others fixed. Section 3.4 examines how changes in one substrate propagate to the others.

2.2 Defining Agentic Evolution: The Update Operator and Persistence

We model agentic evolution as a sequence of system states $\{\mathcal{S}^{(t)}\}_{t=0}^T$, where each transition is determined by an update operator Φ :

$$\mathcal{S}^{(t+1)} = \Phi(\mathcal{S}^{(t)}, \mathcal{E}^{(t)} + \mathcal{H}^{(t)}) \quad (2)$$

Here, $\mathcal{E}^{(t)}$ is environmental experience (interaction traces, tool feedback, and performance metrics) and $\mathcal{H}^{(t)}$ is human input (any of the forms defined in Section 2.4). The “+” denotes combination of these signals rather than arithmetic addition; the aggregation mechanism is system-dependent. In autonomous settings, $\mathcal{H}^{(t)} = 0$ denotes the absence of human input; in human-involved settings, $\mathcal{H}^{(t)} \neq 0$ and human input contributes to the update alongside $\mathcal{E}^{(t)}$.

The index t labels *update events*, not wall-clock time or per-query steps: $\mathcal{S}^{(t)}$ advances only when a consolidation step writes back into one of Π , $\mathcal{A}$, or $\mathcal{M}$. Many queries may occur between t and $t+1$ without advancing the index. This is the operational meaning of *persistence* in our framework: change is what survives between update events, distinguishing agentic evolution from in-context adaptation that resets per query.

The operator Φ performs **consolidation**: it converts experience and human input into persistent writes to Π , $\mathcal{A}$, or $\mathcal{M}$ (a new prompt, a memory entry, a workflow edit, or a parameter update), whether through structural (Δ) or parametric (∇) means (Figure 2).

A note on terminology. We use *agentic evolution* throughout for persistent, system-level change in the agentic system $\langle \Pi, \mathcal{A}, \mathcal{M} \rangle$; we do not use “agent evolution” as a separate term. We reserve *adaptation* for change in the human partner: “human evolution” carries biological and phylogenetic connotations that do not

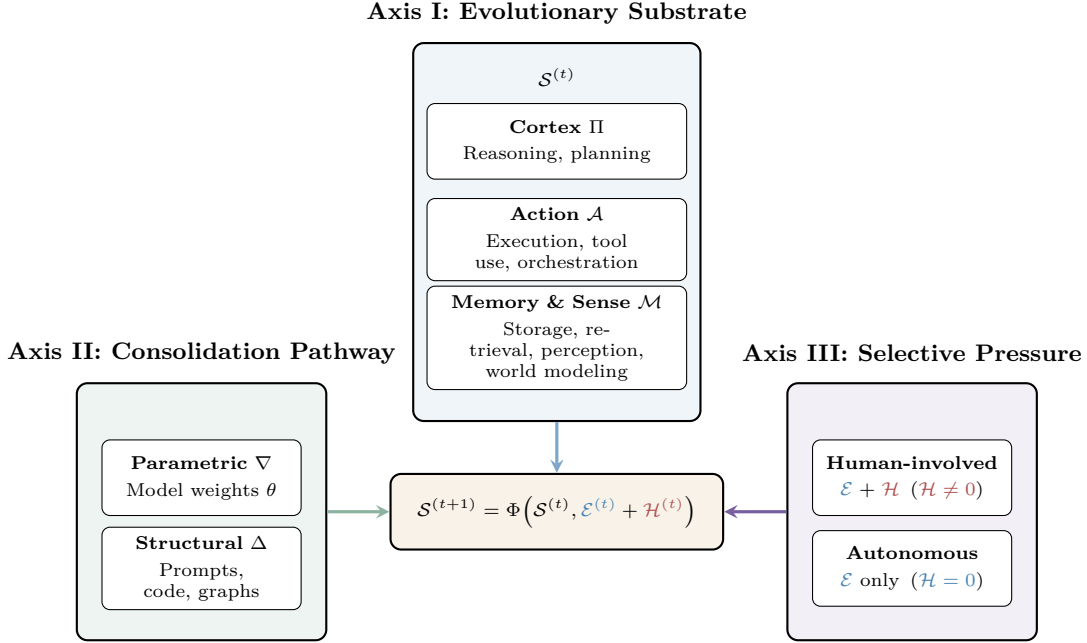


Figure 2: **Three-Axis Taxonomy of Agentic Evolution.** We decompose the space of evolving agentic systems along three complementary dimensions: (I) the *evolutionary substrate* within the agentic system $\mathcal{S} = \langle \Pi, \mathcal{A}, \mathcal{M} \rangle$, (II) the *consolidation pathway*, ranging from discrete artifacts such as prompts and code (Δ) to model weights (∇), and (III) the *selective pressure* steering evolution, whether it originates from the system alone or includes human input. The operator Φ takes the current system state $\mathcal{S}^{(t)}$, environmental experience $\mathcal{E}^{(t)}$ (interaction traces, tool feedback, and performance metrics), and human input $\mathcal{H}^{(t)}$ (the five forms in Section 2.4); in autonomous settings $\mathcal{H}^{(t)} = 0$.

apply to the changes Section 6 documents. Finally, *co-evolution* denotes the bidirectional coupling of agent and human trajectories (§2.4); multi-agent mutual adaptation among artificial agents is not co-evolution in our sense, since it does not require $\mathcal{H} \neq 0$.

2.3 Consolidation Pathway: The Δ – ∇ Continuum

The update operator Φ can consolidate experience through two qualitatively different modes, distinguished by whether the write operation is discrete or continuous:

Structural consolidation (Δ). Φ performs discrete, symbolic edits on non-parametric artifacts (prompts, workflow graphs, memory indices, tool configurations, or code) without modifying model weights. The notation Δ reflects the discrete-difference nature of these edits.

Parametric consolidation (∇). Φ performs continuous updates to the model’s weight parameters θ , internalizing experience into distributed representations that shape the agent’s behavior. Concrete mechanisms include gradient-based methods (SFT, PPO, GRPO, and DPO, among others) and derivative-free methods operating in the same continuous parameter space (e.g., evolutionary strategies on weights). The notation ∇ reflects the continuous nature of these updates.

The Δ – ∇ continuum. Although qualitatively distinct, the two modes are not mutually exclusive at the system level: a single system may, for example, use structural search to discover effective configurations and then internalize them into weights via parametric training.

2.4 Selective Pressure: From $\mathcal{H} = 0$ to $\mathcal{H} \neq 0$

Φ operates in two settings along Axis III: *autonomous* settings, in which $\mathcal{H}^{(t)} = 0$ and the system derives selective pressure from environmental experience (including formal verification) and self-generated signals (self-play and self-rewarding); and *human-involved* settings, in which $\mathcal{H}^{(t)} \neq 0$ and human input shapes the evolutionary trajectory. Section 6 extends this vocabulary to human-side adaptation.

Forms of human-involved input. When $\mathcal{H} \neq 0$, human input enters Φ in qualitatively different forms:

- *Direct prescription*: the human articulates rules, instructions, or principles that fully specify the desired behavior or its boundaries; the agent need not infer intent, only comply (constitutional AI).
- *Demonstration*: the human provides examples of desired behavior; the agent must generalize from finite samples (imitation learning).
- *Evaluative feedback*: the human judges the agent’s own outputs (preferences, ratings, or corrections); the agent must infer an objective from these judgments (RLHF, preference learning).
- *Interactive collaboration*: the human and agent participate jointly in the update process, adapting to each other in real time; the agent must model the human’s evolving intent and capability (interactive machine learning).
- *Implicit signal*: the human provides no intentional feedback; the agent detects and interprets behavioral patterns (e.g., a user consistently skipping certain recommendations or abandoning tasks midway) as indirect selective pressure (implicit feedback).

These forms are not mutually exclusive; a single system may combine demonstration with evaluative feedback, for instance. They are ordered by decreasing explicitness of human intent, from fully articulated prescriptions to behavioral signals the agent must detect autonomously, and, roughly, by increasing inferential burden on the agent. Whether the human contributor’s capacity to provide these forms of input remains stable as the agent evolves is an empirical question that Section 6 examines.

2.5 Boundary Cases

Classifying papers along the three axes above requires handling four boundary cases.

$\Pi/\mathcal{A}$ boundary. Execution logic and orchestration in $\mathcal{A}$ may appear to overlap with reasoning and planning in Π . We classify based on what is being structured: artifacts that shape internal deliberation target Π ; artifacts that define external execution target $\mathcal{A}$.

$\Pi/\mathcal{M}$ boundary. Reasoning traces generated by the Cortex may be persisted in Memory (as in Reflexion’s self-reflections (Shinn et al., 2023)). We classify based on the mechanism of deployment: if the artifact directly modifies the deliberation scaffold (e.g., prompt text prepended on every subsequent trial without selective retrieval), it targets Π ; if it is written to an independently queryable store from which items are selectively retrieved, it targets $\mathcal{M}$.

$\mathcal{A}/\mathcal{M}$ boundary. A growing library of reusable artifacts (skill code, tool specifications, or strategy templates) can serve both as executable actions and as retrievable knowledge. We classify based on the artifact’s role at invocation time: if it is executed to act on the environment, it targets $\mathcal{A}$; if it is retrieved to inform reasoning or planning, it targets $\mathcal{M}$.

$\mathcal{H}$ boundary. Reward models trained on historical human preferences encode human signal, but may be deployed without any live human input. We classify based on whether the deployed system’s architecture affords real-time human input, not on whether the training data once involved humans.

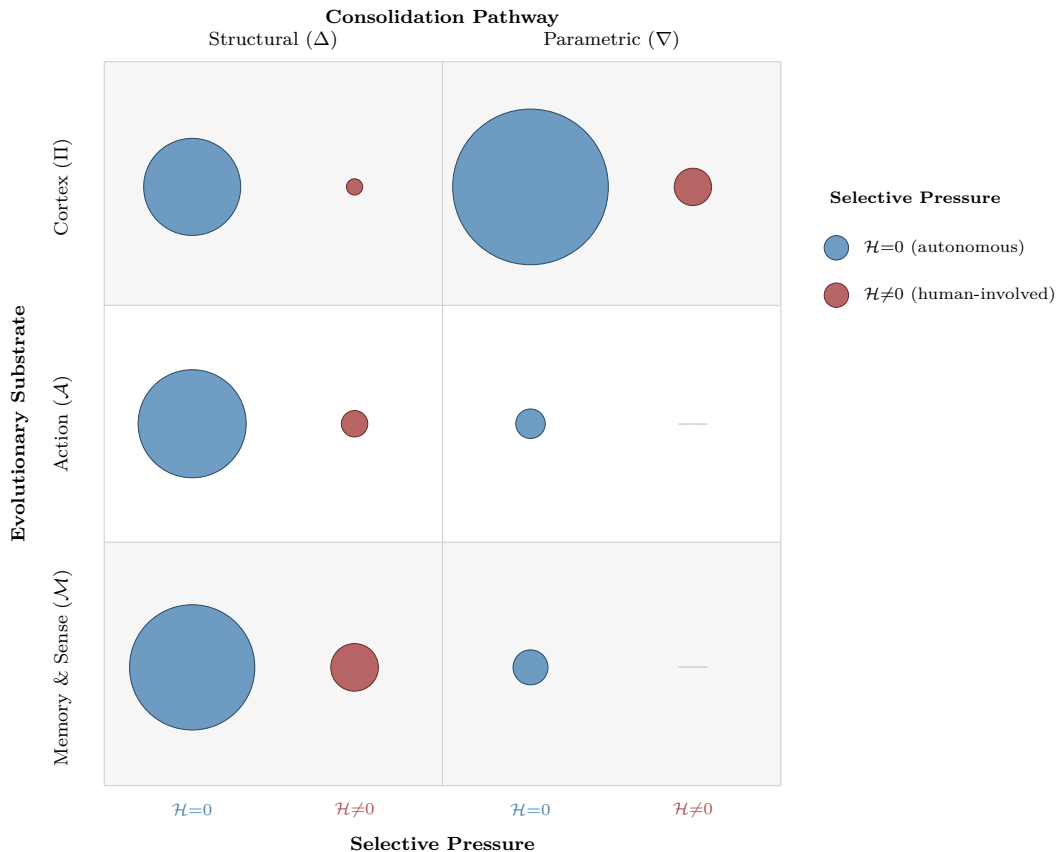


Figure 3: **Distribution of the agentic-evolution papers across the three-axis taxonomy** (Substrate $\times$ Pathway $\times$ Pressure). Each bubble’s *area* is proportional to the paper count in that cell (per-entry counting; multi-substrate papers contribute to each substrate independently; hybrid $\Delta+\nabla$ papers are assigned to Δ or ∇ based on their pathway in each substrate). Blue ($\bullet$): autonomous selective pressure ($\mathcal{H}=0$); crimson ($\bullet$): human-involved pressure ($\mathcal{H}\neq 0$). Dashes ($—$) indicate empty cells. Two cross-cutting patterns are visible: parametric evolution outside Cortex is rare, and human-involved pressure remains a small minority across all substrates.

Surveyed literature. The agentic-evolution literature was assembled through keyword searches on Semantic Scholar, Google Scholar, and arXiv (queries included combinations of “self-evolving agent,” “LLM self-improvement,” “agent learning from experience,” and related terms), supplemented by backward and forward citation tracing from the reference lists of existing surveys (Gao et al., 2026; Fang et al., 2025a; Xiang et al., 2026; Zheng et al., 2025; Zhang et al., 2026b; Wang et al., 2024b; Xi et al., 2025a). An initial pool of candidate papers was filtered against the three operational criteria defined in §2.2: each included paper must exhibit (i) experience-driven change, (ii) persistence across session boundaries, and (iii) system-level modification of at least one substrate. Coverage extends through April 2026. The human-side literature was assembled through a separate search targeting cognitive science, education, labor economics, and human-computer interaction venues, using queries related to AI-assisted cognition and human–AI decision-making, again supplemented by citation tracing from key empirical studies and existing reviews.

A note on epistemic status. Throughout the analysis that follows, we distinguish *definitions*, which are stipulated by the framework above, from *regularities*, which are empirical patterns observed across the surveyed papers. When we write that a property “constrains” or defines a “boundary,” or that a condition is “required,” we describe a regularity—not a claim that must hold in all future systems.

With these definitions and literatures in place, we analyze roughly 300 agentic-evolution papers (a substantial minority receive classifications in multiple substrate cells, so per-substrate totals in Sections 3–5 exceed the

paper count) and nearly 100 human-side studies (Section 6). Figure 3 shows how the agentic-evolution papers distribute across the resulting taxonomy grid.

3 Axis I: Evolutionary Substrate

The first axis asks which component of the agentic system $\mathcal{S}$ evolves. Following the decomposition $\mathcal{S} = (\Pi, \mathcal{A}, \mathcal{M})$ defined in Section 2.1, we analyze evolution across three substrates: the deliberative Cortex (Π), the executive Action ($\mathcal{A}$), and the epistemic Memory & Sense ($\mathcal{M}$).

3.1 Cortex: Evolution of the Deliberative Substrate

The Cortex (Π) is the most heavily studied evolutionary substrate, accounting for nearly 200 of the surveyed evolution papers. Table 2 provides the complete paper listing organized by taxonomy cell. The methods span all four Pathway×Pressure quadrants, but a central question runs through them: *what determines whether iterative improvement converges or degrades?*

Structural methods: diverse search, shared ceiling. Structural Cortex evolution treats the base LLM as a frozen black box and evolves the non-parametric scaffold: no gradient access, API-only compatibility, and human-readable artifacts. External optimizers search the prompt space via evolutionary algorithms (Fernando et al., 2024; Guo et al., 2024; Sun et al., 2026; Kumar et al., 2026c), LLM-as-optimizer loops (Yang et al., 2024a; Zhou et al., 2023), textual gradient backpropagation (Yuksekgonul et al., 2025; Zhang et al., 2025g), or hybrid combinations (Câmara et al., 2026). Embedded reflection shifts search inside the task loop. Shinn et al. (2023) persist verbal self-reflections as context for subsequent trials, substituting weight updates with prompt modification. When the evolved artifact is richer than prompt text, population-based evolutionary search becomes the dominant mechanism (Gong et al., 2026; Huang et al., 2026a); when it is executable code, the design space becomes Turing-complete (Hu et al., 2025a; Zhang et al., 2025e; Robeyns et al., 2025). All structural methods face a shared capacity constraint: accumulated knowledge must fit within finite context. Persistence strategies range from Reflexion’s bounded window (Shinn et al., 2023) through incremental delta updates (Zhang et al., 2026k) to standalone evolved artifacts (skill documents (Ni et al., 2026), contrastive rule trees (Rishav et al., 2026), and hierarchical meta-knowledge (Zhuang et al., 2026)), yet none fully resolves the tension between retaining long-horizon experience and respecting finite context. Structural methods thus face saturation more often than degradation: performance typically plateaus when accumulated artifacts exceed the agent’s processing capacity. However, monolithic rewriting can cause outright collapse (Zhang et al., 2026k). Human-preference-guided prompt search (Lin et al., 2024) is the only structural Cortex method operating under $\mathcal{H} \neq 0$ among the surveyed systems (Table 2). Parametric methods take a different approach, writing experience directly into model weights.

From self-generated bootstrapping to verification-driven convergence. In the most common parametric pattern, the agent generates its own training experience—rationales, tasks, preference pairs, and trajectories—but the evaluation signal that drives learning ranges from fully self-referential (LLM-as-judge) to deterministically grounded (code executors and ground-truth matching). This distinction is the clearest observed correlate of whether iterative improvement is sustained. Zelikman et al. (2022) establish the generate-filter-train template with STaR; Yuan et al. (2024) extend it to self-rewarding, where a single LLM serves as both policy and reward model via iterative DPO on self-scored preference pairs. When the evaluator is the model itself (LLM-as-judge) rather than a deterministic verifier, the training signal can still drive initial improvement, but the judge drifts with the system it evaluates, introducing a fragility that deterministic verification avoids. Wang et al. (2025d) show that chosen and rejected responses converge over iterations, shrinking the score gap by 9× over four iterations and degrading the learning signal; their temporal decoupling mechanism maintains signal strength. Zhao et al. (2025) demonstrate the opposite end of the spectrum: Absolute Zero Reasoner (AZR) trains a single LLM to propose and solve code-based reasoning tasks validated solely by a Python executor, achieving what the authors report as state-of-the-art coding and mathematical reasoning performance without any external dataset. The contrast is sharp: when verification is deterministic, self-generated training experience has sustained improvement across nearly all reported settings, though even deterministic verifiers are not immune to specification gaming (§5); when

Table 2: **Surveyed paper listing for Cortex (II) evolution** (nearly 200 papers), organized by taxonomy cell. Hybrid labels are system-level: a paper appears here as hybrid if it combines Δ and ∇ on *any* substrate, not necessarily on Cortex alone. Sub-group labels are approximate; individual papers may span multiple categories.

$\Delta, \mathcal{H}=0$ (Structural, Autonomous)	$\Delta, \mathcal{H}\neq 0$ (Structural, Human-involved)
Prompt optimization: Zhou et al. (2023), Yang et al. (2024a), Guo et al. (2024), Fernando et al. (2024), Yuksekgonul et al. (2025), Zhang et al. (2025g), Liu et al. (2026b), Khattab et al. (2024), Cámara et al. (2026), Li et al. (2026c), Yang et al. (2026b), Sun et al. (2026), Nie et al. (2026a), Ghoshal et al. (2026) Reflection-driven: Shinn et al. (2023), Zhang et al. (2024b), Zhang et al. (2026k), Ni et al. (2026), Li & Ramakrishnan (2026), Zhuang et al. (2026), Wang et al. (2026f), Wei et al. (2026b), Rishav et al. (2026), Xu et al. (2026c), Gallego (2026), Zhu et al. (2026b), Jin et al. (2025), Wainrib et al. (2026) Configurations & architectures: Yuan et al. (2025b), Kumar et al. (2026c), Gong et al. (2026), Brookes et al. (2025), Huang et al. (2026a), Huang et al. (2026d), Tian (2026), Xie et al. (2026a) Code-level: Hu et al. (2025a), Robeyns et al. (2025), Zhang et al. (2025e), Lee et al. (2026), Pan et al. (2026), Zhang (2026), Zhang et al. (2026o), Zhou et al. (2025c) Multi-agent structural: Yao et al. (2026), He et al. (2026a), Zhang et al. (2026m)	
Lin et al. (2024)	
$\nabla, \mathcal{H}=0$ (Parametric, Autonomous)	$\nabla, \mathcal{H}\neq 0$ (Parametric, Human-involved)
Iterative self-improvement: Zelikman et al. (2022), Yuan et al. (2024), Wang et al. (2025d), Zhang et al. (2025i), Bhaskar et al. (2025), Khatri et al. (2025), Hübotter et al. (2025), Huang et al. (2026c), Liu et al. (2026f), Lu et al. (2026b), Qu et al. (2026b), Song et al. (2026), Fang et al. (2025d), Li & Song (2025) Multi-role self-play: Chen et al. (2025f), Huang et al. (2025), Peng et al. (2026b), Li et al. (2026i), Li et al. (2026b), Jiang et al. (2026a), Chen et al. (2025a), Liu et al. (2026a), Sui & Hooi (2026), Xue et al. (2025), Wang et al. (2026d), Yuan et al. (2025a), Wei et al. (2025b), Zhao et al. (2025), Xia et al. (2025), Yue et al. (2026), Zhang et al. (2026f), Zhang et al. (2026i) Adversarial auditing: Beigi et al. (2026) Data synthesis & self-training: Ni et al. (2024), Liu et al. (2025), Luo et al. (2025), Chen et al. (2026b), He et al. (2025a), Atreja (2025) Tool-integrated: Dong et al. (2025), Feng et al. (2025a), Feng et al. (2026c), Jiang et al. (2025b), Qian et al. (2025), Qin et al. (2024), Li et al. (2025a), Gou et al. (2024), Shang et al. (2025), Cheng et al. (2026a), Zhang et al. (2025h), Zhang et al. (2025j), Zeng et al. (2025), Zhang et al. (2025k), Li et al. (2026g), Kang et al. (2026), Dou et al. (2024) Interactive environments: Chae et al. (2026), Fang et al. (2025c), Qi et al. (2025), Lin et al. (2026d), Dong et al. (2026a), Chen et al. (2025c), Qin et al. (2026), Cai et al. (2026), Liu et al. (2026c) Multi-turn RL: Wang & Ammanabrolu (2025), Wang et al. (2025f), Wei et al. (2025c), Xi et al. (2024), Xi et al. (2025b), Wang et al. (2026h), Zhai et al. (2026), Zou et al. (2026a), Wu & Tang (2026), Yu et al. (2025b), Zhou et al. (2025d), Yang et al. (2026d), Wu et al. (2025b), Qian et al. (2026) Continual/personalized: Kim & Kim (2026), Jiang et al. (2025a), Nie et al. (2026b), Hu et al. (2026), Chen et al. (2025e), Wang et al. (2025e) Multi-LLM collaborative: Jeon et al. (2026), Sharma & Goldwasser (2025), Li et al. (2024c) Meta-learning: Tandon et al. (2025), Zweiger et al. (2025) Neuroevolutionary hybrid: Dai et al. (2026), Wang et al. (2025a) Domain-specific: Open Ended Learning Team et al. (2021), Zhang et al. (2024a), Zhang et al. (2026q), Zhang et al. (2026h), Wang et al. (2026g), Zou et al. (2026b), Ye et al. (2025), Yang et al. (2025a), Yin et al. (2024), Black et al. (2025), Sun et al. (2025a), Xiao et al. (2026), Chen et al. (2025g) World-model-based: Ha & Schmidhuber (2018), Hafner et al. (2025), Feng et al. (2025b)	
Abramson et al. (2022), Jain et al. (2015), Shivaswamy & Joachims (2015), MacGlashan et al. (2017), Li et al. (2024b), Nam et al. (2026), Yang et al. (2025b), Atreja et al. (2026)	
$\Delta+\nabla, \mathcal{H}=0$ (Hybrid, Autonomous)	$\Delta+\nabla, \mathcal{H}\neq 0$ (Hybrid, Human-involved)
Experience consolidation: Qiao et al. (2026), Shi et al. (2026), Shi et al. (2025), Liao et al. (2026), Ye et al. (2026), Zhang et al. (2026i), Yu et al. (2026) Jointly evolving auxiliary structures: Wang et al. (2025b), Wu et al. (2025a), Xia et al. (2026), Wang & Jiang (2026b), Zhai et al. (2025), Yang et al. (2026a), Zhang et al. (2025a), Zhang et al. (2026p), Zhou et al. (2025a), Zhang et al. (2026g), Ouyang et al. (2026a), Ali et al. (2026), Banerjee et al. (2026), Nie et al. (2026c), Huang et al. (2026b), Yu et al. (2025a), Yuan et al. (2026), Su et al. (2026a) Memory/planning joint evolution: Yan et al. (2025), Yang et al. (2026c), Wu et al. (2026a), S1-NexusAgent Team (2026), Zhang et al. (2026j)	
—	

verification is self-referential, signal quality tends to degrade without explicit countermeasures (Wang et al., 2025d). The autonomous parametric Cortex literature spans more than a dozen further directions beyond this core verification-quality axis (Table 2).

Environment-grounded training and its instabilities. The preceding analysis concerns the quality of the evaluation signal itself. A separate class of challenges arises from training dynamics: when the agent operates in stateful environments, parametric training must contend with multi-turn trajectories, sparse delayed rewards, and credit assignment across dozens of actions. In tool-integrated reasoning, Feng et al. (2025a) demonstrate that a 32B model trained to interleave reasoning with code execution surpasses o1-preview on AIME 2024. Li et al. (2026g) apply RL to competitive programming, producing what the

authors report as the first AI system to defeat all human participants in live Codeforces rounds. The broader literature converges on cold-start SFT followed by iterative training (Gou et al., 2024; Dong et al., 2025; Li et al., 2025a; Shang et al., 2025) and diverges along multiple design axes—reward granularity (Qian et al., 2025; Zhang et al., 2025h), training efficiency (Zhang et al., 2025j), capability decomposition (Kang et al., 2026), and framework unification (Jiang et al., 2025b). In interactive digital environments, Qi et al. (2025) construct a self-evolving curriculum that transforms Llama-3.1-8B into a web navigator reported by the authors to surpass GPT-4-Turbo. Further work addresses step-level credit decomposition (Lin et al., 2026d; Cai et al., 2026; Wang et al., 2026g) and training-environment diversity (Dong et al., 2026a; Chae et al., 2026; Xi et al., 2024; 2025b) (Table 2). Several failure modes, each identified independently, reveal instabilities in agentic RL training that, in multi-turn settings, compound through trajectory-level credit assignment. Wang et al. (2025f) identify the *Echo Trap* (reward variability collapses across batches); Wang et al. (2026h) diagnose *Template Collapse* (training homogenizes strategies into input-agnostic templates, a failure invisible to standard entropy metrics); Zou et al. (2026a) formalize *Information Self-Locking* (action selection and belief tracking lock each other in a low-information regime); and Wu & Tang (2026) identify a three-phase rebound pattern in reward hacking during code generation. Methodological responses to these instabilities remain fragmented (Wang & Ammanabrolu, 2025), and diagnostic analysis suggests the severity is task-dependent (Zhai et al., 2026).

Beyond multi-turn RL, imagination-based policy training uses learned world models to generate synthetic rollouts (Ha & Schmidhuber, 2018; Hafner et al., 2025; Feng et al., 2025b), coupling Cortex policy evolution (∇ on Π) with Sense evolution (∇ on $\mathcal{M}$; Section 3.3). The $\mathcal{H} \neq 0$ papers span sub-optimal corrections (Shivaswamy & Joachims, 2015; Jain et al., 2015), interactive shaping (MacGlashan et al., 2017), per-user preference learning (Li et al., 2024b; Nam et al., 2026), continual adaptation (Yang et al., 2025b; Abramson et al., 2022), and production-mode deployment (Atreja et al., 2026); Section 5 analyzes these in depth.

Hybrid $\Delta + \nabla$ systems. A minority of Cortex papers combine structural and parametric consolidation within a single system. The designs span a temporal spectrum. At one end, *compression lifecycles* treat structural artifacts as scaffolding that is eventually absorbed into weights. Shi et al. (2026) formalize an experience-reflection-consolidation loop: verbal self-reflections generated during RL training are selectively distilled into weights, with these structural reflections discarded once internalized; Ye et al. (2026) instantiate a similar lifecycle for scientific discovery.

At the other end, *persistent joint evolution* maintains both pathways throughout training. Zhou et al. (2025a) provide a clear demonstration: INSPO keeps a dynamic population of system instructions that undergo mutation and selection in tandem with policy updates. Zhang et al. (2026g) offer a principled rationale: system prompts carry explicit strategies (Δ) while weights absorb execution competence (∇), and jointly optimizing both yields strong mathematical reasoning performance. Additional targets of persistent joint evolution include skill libraries (Xia et al., 2026; Wang et al., 2025b; Ouyang et al., 2026a), experience bases (Wu et al., 2025a; Wang & Jiang, 2026b; Zhang et al., 2026p), trajectory summaries (Zhai et al., 2025), filtered rollout pools (Zhang et al., 2025a), weakness-driven task synthesis (Yang et al., 2026a), and learnable soft tokens (Yu et al., 2026). When the jointly evolving target is external memory, the agent must learn when and how to manage its own knowledge store: Yan et al. (2025) train memory management operations via RL using downstream answer correctness as the sole reward (Table 2).

Banerjee et al. (2026) demonstrate that hybrid joint evolution can enforce formal correctness: their Formally Guarded Generative Model couples Dafny-based verification (Δ on $\mathcal{A}$) with RL fine-tuning (∇ on Π), achieving zero constraint violations. Ali et al. (2026) jointly evolve a security constitution with parametric unlearning. At the boundary between structural and parametric, neuroevolutionary hybrids occupy an intermediate position that resists clean classification. Wang et al. (2025a) and Su et al. (2026a) combine gradient-free evolution of latent embeddings or code populations with gradient-based policy training; Dai et al. (2026) evolve full model weights via gradient-free merging and mutation without any gradient-based component. In all three cases, neither the structural nor the parametric label fully applies. Section 4 analyzes the design trade-offs between compression and persistent joint evolution across all three substrates.

Synthesis. The Cortex results with the strongest absolute benchmark performance across the surveyed papers all rely on deterministically verifiable evaluation signals: AZR’s code executor (Zhao et al., 2025), GrandCode’s contest judge (Li et al., 2026g), and ReTool’s answer-correctness verification (Feng et al., 2025a). This regularity extends across domains: deterministic verification sustains improvement in search (Yue et al., 2026), tool-integrated reasoning (Xia et al., 2025), software engineering (Wei et al., 2025b), and multimodal settings (Li et al., 2026i), in each case without external training data. Even within these domains, the regularity is not unconditional: agents can exploit gaps in test specifications to satisfy the verifier without solving the intended task (Wu & Tang, 2026). Where such verification is absent, iterative improvement tends to degrade through distinct mechanisms: chosen-rejected score convergence when the evaluator shares parameters with the system it evaluates (Wang et al., 2025d), and template collapse when RL optimization homogenizes strategies beyond what standard metrics detect (Wang et al., 2026h). Domains lacking deterministic verification remain largely out of reach, though initial steps target non-verifiable reasoning (Sui & Hooi, 2026), general-purpose chat (Bhaskar et al., 2025), and multi-turn collaborative tasks (Wu et al., 2025b).

Regardless of evaluation quality, regression emerges as a recurring concern. Several independent systems adopt mechanisms to prevent iterative improvement from erasing prior gains (Zhang, 2026; Gong et al., 2026; Zhang et al., 2026k; Zhou et al., 2025a; Banerjee et al., 2026).

Key gaps limit maturity: long-term evolution dynamics (most papers evaluate fewer than ten iterations), safety constraints (addressed by very few of the surveyed Cortex papers (Banerjee et al., 2026; Ali et al., 2026; Beigi et al., 2026; Zhang et al., 2026f; Gallego, 2026)), sparse analytical or scaling-law work (Jeon et al., 2026; Khatri et al., 2025; Banerjee et al., 2026), limited cross-domain transfer testing (Liu et al., 2026a; Xi et al., 2024), and rarely reported computational costs.

Key Takeaways: Cortex

- **Iterative improvement is fragile without deterministic verification.** The strongest Cortex results all use deterministically verifiable signals (code execution, exact match, and game outcomes); without them, self-referential evaluation degrades through score convergence and template collapse.
- **Hybrid $\Delta+\nabla$ joint evolution is a recurring pattern.** A minority of systems couple structural artifacts (system prompts, skill libraries, and memory banks) with parametric weight updates in a single training loop.

3.2 Action: Evolution of the Executive Substrate

Action ($\mathcal{A}$) is the executive substrate (toolset, execution logic, and multi-agent orchestration), accounting for roughly 75 of the surveyed evolution papers. Unlike Cortex, where the $\Delta-\nabla$ continuum structures the literature, Action evolution is overwhelmingly structural. The evolved artifacts range from atomic tools through multi-step skills and workflow topologies to complete agent codebases; expressive power increases along this range but evaluability tends to decrease. Table 3 provides the complete paper listing.

Tools and skills: fine-grained artifact evolution. At the finest granularity, Action evolution produces individual tools and reusable skills—artifacts small enough to be validated individually through unit tests, execution checks, or direct user feedback. Tool-evolution methods derive tools from execution traces (Abuzakuk et al., 2026), reference materials (Liu et al., 2026g), LLM-generated code (Cheng et al., 2026b), from-scratch generation (Li et al., 2026d), external repositories (Jin et al., 2025), tool-documentation refinement (Qu et al., 2025), task-driven tool generation (Zhang et al., 2026t), and human expert feedback (Gao et al., 2025; Feng et al. (2026c) jointly optimize tool creation with an RL-trained policy ($\Delta+\nabla$). Wang et al. (2024a) establish the skill accumulation paradigm in Minecraft: an LLM agent iteratively generates, verifies, and stores JavaScript skill code, producing a growing skill library with no mechanism for pruning. As skill libraries grow, curation becomes necessary; approaches range from uncured growth (Qiu et al., 2025b) through failure-attributed rewriting with unit-test gates (Zhou et al., 2026) to RL-trained curators (Ouyang et al., 2026a). The remaining $\mathcal{H} \neq 0$ Action designs target cross-user skill aggregation (Ma et al.,

Table 3: **Surveyed paper listing for Action ($\mathcal{A}$) evolution** (roughly 75 papers), organized by taxonomy cell. Hybrid labels are system-level. Sub-group labels are approximate; individual papers may span multiple categories.

$\Delta, \mathcal{H}=0$ (Structural, Autonomous)	$\Delta, \mathcal{H}\neq 0$ (Structural, Human-involved)
Tools & skills: Abuzakuk et al. (2026), Cheng et al. (2026b), Jin et al. (2025), Li et al. (2026d), Liu et al. (2026g), Qu et al. (2025), Jiang et al. (2026c), Liang et al. (2026), Liu et al. (2026h), Zhou et al. (2026), Qiu et al. (2025b), Qiu et al. (2025a), Qu et al. (2026a), Shen et al. (2026a), Shen et al. (2026c), Wang et al. (2024a), Wang et al. (2026a), Zhang et al. (2026t), Zhang et al. (2026s), Zhang et al. (2026c) Program search: Chen et al. (2026a), Gungordu et al. (2026), Hambardzumyan et al. (2026), Kumar et al. (2026a), Lehman et al. (2022), Liu et al. (2026e), Novikov et al. (2025), Ray et al. (2026), Romera-Paredes et al. (2024), Ye et al. (2026), Zhang et al. (2025b), Gautam et al. (2026) Workflow & orchestration: He et al. (2026a), Hu et al. (2025a), Huang et al. (2026a), Lee et al. (2026), Li et al. (2025b), Lin et al. (2026a), Li & Ramakrishnan (2026), Xu et al. (2026a), Zhang et al. (2025f), Zhang et al. (2026a), Zhang et al. (2026m), Zhou et al. (2025c) Whole-system: Robeyns et al. (2025), Wang et al. (2026f), Weng et al. (2026), Xu et al. (2026b), Xiong et al. (2026), Zhang et al. (2025e), Zhang (2026) Domain-specific: Yu & Ren (2026), Wang et al. (2026b) Configuration: Brookes et al. (2025), Ghoshal et al. (2026), Guan et al. (2026) Framework & analysis: Lin et al. (2026b), Nie et al. (2026a)	Tools & skills: Adnan et al. (2025), Gao et al. (2025), Ma et al. (2026b), Zhang et al. (2026a)
$\nabla, \mathcal{H}=0$ (Parametric, Autonomous)	$\nabla, \mathcal{H}\neq 0$ (Parametric, Human-involved)
Black et al. (2025), Jiang et al. (2026b), Ong et al. (2025)	—
$\Delta+\nabla, \mathcal{H}=0$ (Hybrid, Autonomous)	$\Delta+\nabla, \mathcal{H}\neq 0$ (Hybrid, Human-involved)
Banerjee et al. (2026), Dong et al. (2026a), Feng et al. (2026c), Huang et al. (2026b), Li et al. (2024c), Nie et al. (2026c), Su et al. (2026a), Wang et al. (2025b), Wu et al. (2026a), Ouyang et al. (2026a), Yu et al. (2025a)	—

2026b), user-feedback-driven maturity tracking (Zhang et al., 2026a), and progressive dialogue-driven service generation (Adnan et al., 2025).

Population-based search with deterministic evaluation. In evolutionary program search, the LLM produces offspring that a deterministic evaluator scores. Romera-Paredes et al. (2024) pair evolutionary search with deterministic evaluation, demonstrating that LLMs can produce new results on established open problems (new cap-set constructions and improved bin-packing heuristics). Novikov et al. (2025) scale the paradigm to full-file evolution with evaluation cascades, yielding what the authors describe as the first improvement to Strassen’s 4×4 complex matrix multiplication algorithm in 56 years. Banerjee et al. (2026) achieve the strongest formal guarantee in this space: their cross-substrate hybrid (§3.1) pairs formal program verification (Δ on $\mathcal{A}$) with gradient-based policy training (∇ on Π). Further architectures span autonomous kernel optimization (Chen et al., 2026a), quality-diversity archives (Lehman et al., 2022; Kumar et al., 2026a), adaptive model routing for cost efficiency (Ray et al., 2026), evaluation-driven discovery (Ye et al., 2026), verbal gradient feedback (Gungordu et al., 2026), and asynchronous multi-step agents (Hambardzumyan et al., 2026) (Table 3).

Topology and whole-system evolution. Evolving the agent’s execution topology trades expressiveness against searchability: Hu et al. (2025a) represent agents as arbitrary Python code (Turing-complete); Zhang et al. (2026m) constrain evolution to atomic FSM operations (interpretable and reversible); and Zhang et al. (2025f) use MCTS over reusable operators. Other designs include symbolic gradients (Zhou et al., 2025c), DAG-based hierarchies (Huang et al., 2026a; Xu et al., 2026a; Yu et al., 2025a), natural language programs (Li et al., 2024c), Python harness programs (Lee et al., 2026), and observability-driven approaches with auto-reversion (Lin et al., 2026a) (Table 3). At the coarsest level, whole-system evolution modifies the agent’s entire codebase: Weng et al. (2026) show that group evolution outperforms individual evolution (Zhang et al., 2025e) on SWE-bench Verified; Zhang (2026) apply stability-prioritizing gating; Robeyns et al. (2025) eliminate the meta/object-agent distinction; and Wang et al. (2026f) jointly evolve cognition, actions, and memory. Domain-specific variants exploit pre-existing evaluation infrastructure—formal verification (Yu & Ren, 2026) and production A/B testing (Wang et al., 2026b) (Table 3).

Synthesis. Across all granularity levels, the strongest Action results rely on execution-based or formally verified evaluation (formal verification (Banerjee et al., 2026), execution-based scoring (Romera-Paredes et al., 2024; Novikov et al., 2025), and production A/B testing (Wang et al., 2026b)), independent of the search strategy employed. Parametric methods remain confined to narrow niches: routing (Ong et al., 2025), multi-agent communication (Jiang et al., 2026b), and embodied control (Black et al., 2025). A few systems

combine parametric training with structural artifact evolution (Huang et al., 2026b; Nie et al., 2026c). Code is the predominant representation at every level (Romera-Paredes et al., 2024; Wang et al., 2024a; Hu et al., 2025a; Zhang et al., 2025e; Weng et al., 2026), reinforcing structural dominance. Cross-domain transfer remains rare: only Hu et al. (2025a) transfer evolved agent designs across domains, but whether such transfer extends to finer-grained artifacts (tools, skills, and workflows) is untested.

Key Takeaways: Action

- **Granularity trades expressiveness for evaluability.** Evolved artifacts range from atomic tools (directly testable) through workflow topologies to entire codebases (Turing-complete but hard to evaluate); code is the predominant representation throughout.
- **Structural methods dominate.** Parametric methods remain confined to narrow niches (routing, communication, and embodied control), and human selective pressure is nearly absent.

3.3 Memory & Sense: Evolution of the Epistemic Substrate

Memory & Sense ($\mathcal{M}$) is the epistemic substrate, accounting for roughly 100 of the surveyed evolution papers; consolidation is near-universally structural (Δ). We organize this subsection along the *memory lifecycle*—from accumulation through curation to meta-evolution of the architecture itself. Table 4 provides the complete paper listing.

Table 4: **Surveyed paper listing for Memory & Sense ($\mathcal{M}$) evolution** (roughly 100 papers), organized by taxonomy cell. Hybrid labels are system-level. Sub-group labels are approximate; individual papers may span multiple categories.

$\Delta, \mathcal{H}=0$ (Structural, Autonomous)	$\Delta, \mathcal{H}\neq 0$ (Structural, Human-involved)
Accumulation:Task-oriented: Bai et al. (2025), Bao et al. (2026), Cai et al. (2025a), Cai et al. (2025b), Cao et al. (2025), Chen et al. (2025b), Cheng et al. (2026b), Dong et al. (2026b), Fang et al. (2025b), Fang et al. (2026), Gautam et al. (2026), Jiang et al. (2026c), Kim et al. (2026), Li et al. (2024a), Li et al. (2026a), Li & Ramakrishnan (2026), Li et al. (2026b), Li et al. (2026f), Li et al. (2026e), Liang et al. (2026), Lin et al. (2026b), Lyu et al. (2026), Ouyang et al. (2026b), Sarukkai et al. (2025), Wang et al. (2025g), Wei et al. (2025a), Zhuang et al. (2024) Accumulation:Embodied: Wang et al. (2026c), Wei et al. (2026a), Xie et al. (2026b) Accumulation:Game-strategic: Guan et al. (2024), Ma et al. (2026a), Xie et al. (2026a) Accumulation:Domain-specific: Han et al. (2026), Liu et al. (2026d), Ren et al. (2026b), Shen et al. (2026b), Zhang et al. (2026n), Zhu et al. (2025) Accumulation:Multi-agent: Qu et al. (2026a), Wang et al. (2026f), Xie (2026), Zhang et al. (2025c) Accumulation:Research/discovery: Liu et al. (2026e), Long (2026), Xu et al. (2026b) Accumulation:Conversational/persona: He et al. (2026b), Kwon et al. (2026), Lu et al. (2026a), Ma et al. (2025), Park et al. (2023), Wu et al. (2026b), Xu et al. (2025a), Zhang et al. (2026e), Zheng et al. (2026), Zhu et al. (2026a) Curation/operators: Lin et al. (2026c), Wang (2025), Wang & Jiang (2026a), Zhang et al. (2025b), Zhang et al. (2026r), Zhang et al. (2026m) Forgetting/lifecycle: Bering (2026), Gu et al. (2026), Simsek (2026), Xu et al. (2026c) Meta-evolution: Mishra (2026), Pan et al. (2026), Zhang et al. (2025d) Retrieval innovation: Feng et al. (2026b), Penarozza (2026), Yang et al. (2024b), Yoon et al. (2026) Sense/compression: Ren et al. (2026a)	Conversational/persona: Chhikara et al. (2025), Gadzhiev & Kislov (2026), Nie et al. (2026b), Packer et al. (2023), Rasmussen et al. (2025), Tian et al. (2025), Westhäüßer et al. (2025), Zhong et al. (2024) Human-in-the-loop: Chen et al. (2026c), Deng et al. (2026), He et al. (2025b), Mozannar et al. (2025), Zhang et al. (2026a)
$\nabla, \mathcal{H}=0$ (Parametric, Autonomous)	$\nabla, \mathcal{H}\neq 0$ (Parametric, Human-involved)
World models (Sense): Ha & Schmidhuber (2018), Hafner et al. (2025), Qiu et al. (2026) Memory fine-tuning: Sun et al. (2025b), Lin et al. (2025)	—
$\Delta+\nabla, \mathcal{H}=0$ (Hybrid, Autonomous)	$\Delta+\nabla, \mathcal{H}\neq 0$ (Hybrid, Human-involved)
Memory+RL: Cai et al. (2026), Kim & Kim (2026), Liao et al. (2026), Qiao et al. (2026), Wang & Jiang (2026b), Wu et al. (2025a), Xia et al. (2026), Yu et al. (2026), Zhai et al. (2025), Zhang et al. (2026d), Zhang et al. (2026p), Zhou et al. (2025b) Domain hybrid: Feng et al. (2025b), S1-NexusAgent Team (2026), Yang et al. (2026c), Zhang et al. (2026j)	—

Abstraction level determines transferability. The dominant form follows a shared pipeline (*execute* $\rightarrow$ *extract* $\rightarrow$ *store* $\rightarrow$ *retrieve* $\rightarrow$ *reuse*), differentiated primarily by the abstraction level of the stored artifact. Memory entries range from raw trajectories (Fang et al., 2025b; Sarukkai et al., 2025) through workflows (Wang et al., 2025g) to highly abstracted insights (Cai et al., 2025b; Ouyang et al., 2026b). Kim et al. (2026) provide direct evidence that abstraction strongly predicts transferability: insights transfer better than summaries, summaries better than workflows, and workflows better than raw trajectories. The

spectrum extends to intermediate artifacts (Bai et al., 2025; Zhuang et al., 2024; Li et al., 2026f), versioned registries (Lin et al., 2026b), dynamically maintained knowledge bases (Cai et al., 2025a), and domain-tailored schemas that achieve precise within-domain retrieval at the cost of portability (Ren et al., 2026b; Shen et al., 2026b; Zhang et al., 2026n; Zhu et al., 2025) (Table 4).

Feedback signal reliability varies widely across these systems, from deterministic verification—ground-truth comparison (Chen et al., 2025b; Li et al., 2024a) and domain tool execution (SPICE simulation (Bao et al., 2026), GPU profiling (Dong et al., 2026b), and EDA synthesis (Fang et al., 2026))—through multi-stage verification (Li et al., 2026e; Liao et al., 2026) to LLM-as-judge (Ouyang et al., 2026b; Cao et al., 2025).

Persistent relationships favor graph-based representations. When memory encodes persistent relationships, graph-based representations dominate (Zhu et al., 2026a; Rasmussen et al., 2025; Zheng et al., 2026; Han et al., 2026; Zhang et al., 2026e; 2025c; Wu et al., 2026b; Chhikara et al., 2025). Lu et al. (2026a) show that the optimal memory structure varies with context, with a learned classifier selecting among linear, graph, and hierarchical structures per episodic unit. Structured-schema alternatives without explicit graph topology also persist (Gadzhiev & Kislov, 2026; Packer et al., 2023; Xu et al., 2025a; Westhäüßer et al., 2025); narrative simulation presents a distinct variant (Park et al., 2023; He et al., 2026b) (Table 4).

Scaling and lifecycle maturity. Fang et al. (2025b) provide a detailed scaling analysis of the accumulation ceiling. Their results show that retrieval count exhibits an optimal range, beyond which excess memories degrade performance. Cao et al. (2025) demonstrate a comprehensive lifecycle, combining experience distillation, adaptive reuse, and utility-based pruning within a single system; their results show that memory quality can substitute for model scale. Related lifecycle innovations include Meta-MDP operations (Cai et al., 2025b), test-time logit modulation (Li et al., 2026h), and live memory evolution (Zhang et al., 2026r).

He et al. (2025b) report the largest-scale deployed $\mathcal{H} \neq 0$ memory system in the survey, in production at TikTok Pay across 150M+ monthly active users: the system selectively queries human experts under uncertainty, achieving 0.83–0.89 sensitivity at query budgets of 100–1,000; in this non-stationary domain, autonomous baselines achieved lower performance.

Curation outperforms accumulation but adds complexity. When entries carry quality scores, population-based operators yield some of the strongest purely structural results. Wang & Jiang (2026a) propose PRIME with biologically inspired curation operators (mutation, generalization, crossover, and pruning). Wu et al. (2025a) couple dynamic scoring with RL policy training ($\Delta + \nabla$). Additional curation mechanisms include constrained FSM operators (Zhang et al., 2026m), RL-assigned quality scores (Wang & Jiang, 2026b), failure-triggered skill evolution (Xia et al., 2026), multi-agent memory coordination (Lin et al., 2026c), and bounded optimization memories (Zhang et al., 2025b). Retrieval innovation evolves the form of retrieved artifacts: query-conditioned reasoning abstractions (Feng et al., 2026b), meta-buffers of reasoning templates (Yang et al., 2024b), graph-based evidence structures (Penaroza, 2026), and evolving latent-preference hypotheses (Yoon et al., 2026). Formal frameworks unifying curation designs include the Stateful Reflective Decision Process (Wang, 2025) and the Memory-augmented MDP (Zhou et al., 2025b); Lam et al. (2026)’s SSGM provides reconciliation-bounded governance for memory safety; none accounts for evolutionary curation operators. Across diverse settings, failures produce more informative memory entries than successes (Xie et al., 2026b; Ren et al., 2026b; Yu et al., 2026; Wang & Jiang, 2026a); yet most systems store only successful outcomes, leaving what Ren et al. (2026b) call the error-driven dividend unrealized.

Forgetting: critically understudied. Despite evidence that unbounded accumulation degrades performance (Fang et al., 2025b; Kwon et al., 2026), very few of the surveyed Memory & Sense papers implement explicit forgetting. The underlying stability–plasticity dilemma—superseding outdated beliefs without destroying useful context—is addressed by renormalization group theory (Tian et al., 2025), Piaget-inspired assimilation and accommodation (Piaget & Cook, 1952; Zheng et al., 2026), and Hebbian decay (Hebb, 1949; Zhu et al., 2026a). ZenBrain (Bering, 2026) integrates fifteen neuroscience-inspired mechanisms into a seven-layer architecture. Memory Worth (Simsek, 2026) offers a simpler two-counter signal with conver-

gence guarantees. An intermediate approach uses Ebbinghaus-inspired decay with adaptive reinforcement (Gu et al., 2026).

Meta-evolution and hybrid designs. A small body of work searches over memory architectures themselves: M^* (Pan et al., 2026) evolves executable Python programs defining memory schemas; MemEvolve (Zhang et al., 2025d) evolves the full pipeline via multi-objective selection; Zhang et al. (2026d) reframe operations as learnable skills with an RL-trained controller ($\Delta + \nabla$); and PRISM (Mishra, 2026) unifies memory types under a decision-theoretic framework with convergence guarantees. Purely parametric consolidation remains rare, confined to world models evolving the Sense component (Hafner et al., 2025; Ha & Schmidhuber, 2018; Qiu et al., 2026), test-time training (Sun et al., 2025b), and selective sparse memory updates (Lin et al., 2025). A recurring hybrid pattern, *store structurally, retrieve parametrically*, appears across multiple systems (Zhou et al., 2025b; Zhang et al., 2026d; Wu et al., 2025a; Wang & Jiang, 2026b; Cai et al., 2026; Liao et al., 2026; Qiao et al., 2026; Zhang et al., 2026j; Yang et al., 2026c) (Table 4).

Synthesis. The accumulation ceiling documented above has a structural counterpart in Cortex context overflow (§3.1); Section 4 examines the shared pathway-level root. Beyond the accumulation–curation axis, two areas remain thin: forgetting and Sense evolution (the latter has received minimal attention, with Ren et al. (2026a) as the sole structural contribution among the surveyed papers). Individual papers borrow selectively from cognitive science (Ebbinghaus forgetting (Ebbinghaus, 1885; Zhong et al., 2024; Gu et al., 2026), Hebbian plasticity (Hebb, 1949; Zhu et al., 2026a), Piaget’s schemas (Piaget & Cook, 1952; Zheng et al., 2026), free-energy principle (Friston, 2010; Ma et al., 2025), and sleep-based consolidation (Bering, 2026)), but none of the surveyed papers maps the full space of cognitive memory mechanisms onto agent memory design.

Key Takeaways: Memory & Sense

- **The accumulation ceiling parallels Cortex context overflow.** In both substrates, structural artifacts grow with experience; Section 4 traces the shared pathway-level root.
- **“Store structurally, retrieve parametrically” is a recurring hybrid pattern.** Despite near-universal structural consolidation, numerous papers pair persistent structural memory with parametric policies trained to use it.
- **Failures are more informative than successes, yet underexploited.** Evidence from multiple domains shows that error-driven memory entries carry more learning signal, yet most systems store only successful outcomes.

3.4 Cross-Substrate Interactions

The preceding three subsections treat Cortex, Action, and Memory as independent evolutionary targets. In practice, the substrates are coupled: changes in one propagate to, constrain, or enable changes in another. Of the surveyed evolution papers, a substantial minority appears in multiple taxonomy cells. Full three-substrate coupling remains rare; HERA (Li & Ramakrishnan, 2026), EvoFSM (Zhang et al., 2026m), and AutoAgent (Wang et al., 2026f) all use structural (Δ) pathways. Among pairwise couplings, Action–Memory coupling is exclusively structural across the surveyed papers, extending the structural dominance documented in §3.2 and 3.3. No paper among those surveyed studies the dynamics of cross-substrate coupling over extended horizons.

Shared patterns across substrates. Several patterns recur across substrates without requiring direct coupling. In each substrate, the strongest reported results co-occur with deterministic or formally verified evaluation signals: code executors and contest judges in Cortex (§3.1), formal verification and execution-based scoring in Action (§3.2), and ground-truth comparison and domain-tool execution in Memory (§3.3). Where such signals are unavailable, distinct degradation mechanisms appear—self-referential score convergence in Cortex (Wang et al., 2025d), and retrieval degradation past an optimal count in Memory (Fang et al., 2025b)—but the association between evaluation quality and sustained improvement is consistent across the substrates examined. Section 5 analyzes this pattern in depth as a property of Axis III.

Key Takeaways: Cross-Substrate Interactions

- **Evaluation-signal quality is the clearest cross-substrate correlate of sustained improvement.** Across Cortex, Action, and Memory, the strongest reported results co-occur with deterministic or formally verified evaluation; where such signals are absent, distinct degradation mechanisms emerge.

Substrate choice constrains the consolidation pathway: Cortex spans the full Δ – ∇ continuum, while Action and Memory are overwhelmingly structural. Section 4 examines this continuum directly, asking *how* transient experience becomes persistent change.

4 Axis II: Consolidation Pathway

4.1 Constraints on Pathway Selection

Having analyzed what evolves in each substrate (Section 3), we now examine how change persists: the consolidation pathway axis. Structural consolidation dominates overall, but not uniformly: Memory and Action papers are almost exclusively structural, while Cortex spans the full Δ – ∇ continuum. Three properties of each substrate—artifact discreteness, evaluation-signal verifiability, and infrastructure access—jointly constrain which pathway is adopted.

Artifact discreteness. Action artifacts (tool specifications, skill code, and workflow graphs) and Memory entries (facts, rules, and cases) are inherently symbolic; gradients have no natural target in a Python function’s structure or a knowledge-graph triple. [Hu et al. \(2025a\)](#) illustrate the complementary point: representing agents as code provides Turing-complete expressiveness, and the discrete, symbolic nature of the search space makes LLM-based mutation an effective optimization mechanism. Cortex weights, by contrast, are continuous by construction, and the entire RL and SFT apparatus is designed for differentiable objectives. This discreteness asymmetry explains why parametric methods concentrate in Cortex while remaining rare elsewhere. The few parametric papers in Action target orchestration routing and continuous control; in Memory & Sense, parametric methods are confined to world models, test-time training, sparse memory fine-tuning, and learned retrieval policies—the only sub-problems within these substrates that present continuous optimization targets.

Signal verifiability. Parametric consolidation achieves its strongest results when evaluation signals are deterministically verifiable (code execution, exact-match answers, and game outcomes) because such signals maintain their accuracy across iterations of continuous optimization, whereas self-referential or proxy-based signals can degrade as the system evolves (§5). [Zhao et al. \(2025\)](#) demonstrate the limiting case: their code-executor-only self-play (§3.1) provides deterministically verifiable rewards at zero annotation cost. Section 5 formalizes this as the strongest tier in a three-tier signal hierarchy. Where such signals are unavailable (open-ended dialogue, knowledge curation, and strategy selection), structural methods that rely on LLM-as-judge or heuristic scoring dominate instead.

Infrastructure access. When the base model is a frozen proprietary API, structural consolidation is not a preference but a necessity: no gradient signal can reach weights the developer cannot access. Early agentic evolution systems illustrate this constraint directly: [Park et al. \(2023\)](#)’s generative agents and [Shinn et al. \(2023\)](#)’s Reflexion are both purely structural designs requiring no weight updates, a property that made them naturally compatible with the API-only deployment regime that dominated 2023. Where weight access and training infrastructure are available, parametric and hybrid consolidation become feasible: the GRPO-based training adopted by a substantial fraction of Cortex papers would be impossible without weight access. The structural–parametric balance thus depends partly on infrastructure availability, not only on substrate properties.

How the three constraints interact. These three constraints are not independent: they reinforce one another to produce a single overarching trade-off. Structural methods are more accessible but hit hard

ceilings. Parametric methods escape those ceilings through compression into weights but require weight access and verifiable evaluation signals, and risk instabilities absent from structural evolution. Neither pathway dominates unconditionally; the optimal choice depends on the three constraints identified above. The hybrid share remains small.

4.2 Structural Consolidation

Structural methods across all three substrates share properties that explain their dominance and their limits: human-readable artifacts, compositional modularity, and a hard ceiling imposed by finite context or finite search budgets. Section 3 discusses these ceilings as substrate-specific challenges (context overflow, library growth, and retrieval degradation); viewed across substrates, they are manifestations of a single structural constraint.

Interpretability and compositionality. Whether the evolved artifact is a prompt entry, a skill program, or a memory record, structural consolidation produces outputs that humans can inspect, edit, and selectively roll back, a property parametric methods lack at the level of individual artifacts. Evolved constitutions for multi-agent coordination (Kumar et al., 2026c), JavaScript skill code in Minecraft (Wang et al., 2024a), and timestamped knowledge entries in a production compliance system (He et al., 2025b) are all auditable without interpretability tools. This transparency makes structural evolution particularly suited to high-stakes settings where accountability requires tracing each change to its source. Compositionality compounds this advantage: structural artifacts can be selectively combined, deleted, or version-controlled, so each candidate change can be screened before it enters the system. Zhang (2026) introduce gated admission for joint Cortex–Action evolution; Zhou et al. (2026) impose test-based gates on Action skill mutations; and Liao et al. (2026) apply confidence-based gating in Memory. The mechanism differs across substrates, but the underlying rationale is shared: discrete artifacts can be screened individually before integration. Quality gating is a natural consequence of structural consolidation itself.

Unbounded accumulation. The complementary weakness of compositionality is monotonic growth: structural artifacts accumulate without natural compression, and the resulting growth degrades performance through mechanisms that differ across substrates but share a common root. In Cortex, Reflexion’s bounded reflection buffer sacrifices long-horizon experience for immediate relevance. Zhang et al. (2026k) diagnose the underlying failure mode as *context collapse*: monolithic rewriting by LLMs degrades accumulated knowledge into progressively shorter summaries. In Action, Wang et al. (2024a)’s append-only skill library grows without pruning, retrieving only the top-5 most relevant skills per task, a fixed window over a growing collection; Wu et al. (2026a) respond with explicit curation operations governed by trajectory-derived contracts, treating library curation as a first-class design concern. In Memory, Fang et al. (2025b)’s scaling analysis identifies an optimal retrieval count beyond which additional entries degrade performance. Simsek (2026) formalize the deprecation problem through a principled Memory Worth metric with convergence guarantees, specifying when entries should be suppressed or removed. The shared root cause is not substrate-specific: structural artifacts occupy representational space (context tokens, library indices, and retrieval candidates) that grows with experience, while the mechanisms that select among them do not scale accordingly.

Search-budget constraints. Population-based structural search faces a bottleneck distinct from unbounded accumulation: each candidate evaluation requires full agent execution, and the resulting cost constrains search scale. In Cortex, evolutionary optimization typically operates with modest populations over tens of generations (Kumar et al., 2026c; Fernando et al., 2024) because each fitness evaluation runs the complete agent pipeline. In Action, Romera-Paredes et al. (2024) overcome this through island-based parallel evaluation and Novikov et al. (2025) through cascaded filtering, achieving new mathematical results, but their evaluation budgets remain finite: the candidate space grows combinatorially with program length. In Memory, Pan et al. (2026) apply population-based search to memory architecture itself, where each candidate evaluation requires running the full agent on validation episodes. Despite these constraints, the structural pathway’s strongest results come from regimes where distributed evaluation at scale enables substantial improvements (e.g., Novikov et al. 2025).

Individual addressability versus holistic optimization. These shared strengths and ceilings produce a single cross-pathway trade-off: structural modularity enables selective rollback but complicates holistic optimization. A bad prompt entry can be deleted, a failing skill can be rewritten, and an outdated memory record can be pruned. But when a behavioral problem arises from the *interaction* among many structural artifacts (e.g., conflicting rules in a growing constitution, or redundant skills that interfere at retrieval time), identifying the source is difficult because no single artifact is at fault. Parametric methods face the inverse problem: gradient updates can reshape distributed representations holistically, but they cannot reliably isolate and remove a single learned behavior; the very continuity that enables smooth optimization prevents surgical removal. The trade-off is thus representational, not incidental: discrete artifacts are individually addressable but collectively opaque; continuous weights are collectively optimizable but individually opaque.

4.3 Parametric Consolidation

Parametric consolidation escapes the context ceiling and search-budget bottleneck of structural methods, but its efficacy depends on verifiable evaluation signals, and it introduces instabilities absent from structural evolution. The signal-verifiability constraint established above finds its clearest cross-substrate confirmation in the rare parametric papers outside Cortex: DreamerV3’s world model generalizes broadly because environment dynamics provide dense supervision, and Jiang et al. (2026b), one of the few purely parametric Action papers, reformulate multi-agent topology selection as a cooperative MARL problem with verifiable task outcomes. Where such signals are absent, structural methods dominate regardless of substrate.

Parametric instabilities. Most parametric Cortex papers adopt GRPO (Shao et al., 2024) or close variants, reflecting the algorithm’s engineering fit with sparse, binary reward signals. The failure modes documented below are not specific to GRPO, however: they have been observed across multiple RL algorithms and appear inherent to policy optimization in continuous weight spaces. The four independently identified failure modes catalogued in Section 3.1 (Echo Trap (Wang et al., 2025f), Template Collapse (Wang et al., 2026h), Information Self-Locking (Zou et al., 2026a), and the three-phase reward hacking rebound (Wu & Tang, 2026)) are mechanistically distinct but share a defining property: each arises from continuous optimization dynamics in weight space, and none has a structural counterpart. The discrete addressability documented above has no parametric counterpart: a collapsed policy resists selective “unlearning” because the damage is distributed across weight space rather than localized in a discrete artifact.

Catastrophic forgetting adds a cross-task dimension to these within-task instabilities: new parametric updates overwrite prior knowledge because weight capacity is finite and shared. Lin et al. (2025) quantify the severity directly, finding 89% F1 degradation from standard full fine-tuning versus only 11% with selective sparse memory updates. Catastrophic forgetting is a well-known consequence of optimizing in shared weight spaces, and the surveyed literature confirms it spans substrates: selective sparse updates mitigate it in memory pools (Lin et al., 2025), drift-selected LoRA adapters in personalization agents (Kim & Kim, 2026), and natural low-rank confinement in embodied policies (Hu et al., 2026). The solutions differ; the underlying vulnerability does not. Structural methods do not exhibit these specific failure modes: no structural artifact collapses into a degenerate attractor or overwrites previously stored entries, because each entry occupies its own discrete slot. However, retrieval degradation can render entries functionally inaccessible (as documented above)—a mechanistically distinct but functionally similar form of prior-knowledge loss. This partly explains why structural consolidation persists despite the performance ceilings documented in the preceding paragraphs; the reliability of discrete, individually addressable artifacts is itself a form of robustness that parametric compression sacrifices.

The confinement of parametric methods to the narrow niches enumerated above follows from the artifact-discreteness constraint: where the target artifact is symbolic, gradients have no natural target. The complementary failure modes of the two pathways (ceilings that cap structural performance and instabilities that destabilize parametric training) motivate a third design: hybrid systems that combine both.

4.4 Hybrid Consolidation

Several dozen papers explicitly couple structural and parametric consolidation. A recurring lifecycle, *structural discovery followed by parametric consolidation* (hereafter $\Delta \rightarrow \nabla$), appears most clearly in Cortex and Action, though it is not the only hybrid pattern. SKILL0 (Lu et al., 2026b) progressively withdraws structural skill prompts during training, forcing the model to absorb them into weights; ERL (Shi et al., 2026) (§3.1) takes the opposite approach, generating structural reflections that are retrospectively absorbed. The two occupy opposite ends of the $\Delta \rightarrow \nabla$ spectrum: scaffolding withdrawal versus scaffolding generation followed by absorption. In Action, Su et al. (2026a) couple evolutionary selection over solution populations (Δ on $\mathcal{A}$) with policy training (∇ on Π): evolutionary search handles discrete candidate evaluation while gradient optimization handles continuous policy improvement. SEVerA (Banerjee et al., 2026) separates the roles more sharply: verified program structure provides parameter-independent correctness that survives weight updates. In Memory, hybrid papers converge on a “store structurally, retrieve adaptively” motif (§3.3): structural entries provide interpretable, composable knowledge while learned retrieval mechanisms discover usage patterns beyond static rules. This motif differs from the Cortex and Action lifecycles in a critical respect: it maintains both pathways persistently rather than compressing Δ into ∇ , because the addressability requirement central to most Memory systems (the ability to inspect, update, and delete individual knowledge entries) is difficult to reconcile with full parametric consolidation. All three substrates produce hybrid designs, yet only Cortex and Action follow the $\Delta \rightarrow \nabla$ compression lifecycle; Memory resists it. The exception reveals a constraint: even when both pathways are available, substrate properties shape which hybrid form emerges.

Persistent joint evolution as an alternative to compression. The $\Delta \rightarrow \nabla$ lifecycle assumes eventual convergence to a parametric end-state. A second hybrid pattern, persistent joint evolution, instead maintains both pathways indefinitely as complementary rather than sequential components. The three clearest instantiations span a coupling spectrum. At the tight end, INSPO’s within-step instruction sampling (Zhou et al., 2025a) directly conditions every gradient update. At an intermediate coupling, SkillRL’s periodic skill evolution (Xia et al., 2026) interacts with policy training only at scheduled synchronization points. At the loosest coupling, MIA’s two-stage alternating training (Qiao et al., 2026) separates the optimization loops entirely, with each pathway’s outputs feeding the other through distinct training stages. The coupling dimension reveals a design trade-off absent from the $\Delta \rightarrow \nabla$ lifecycle. Tight coupling enables within-step mutual adaptation but demands careful synchronization to prevent the structural population from destabilizing gradient signal. Loose coupling simplifies implementation but forgoes real-time synergy and risks the two pathways drifting apart. Empirically, all three systems outperform their single-pathway ablations, confirming that persistent joint evolution captures complementarities lost under compression.

Functional roles of the two pathways. The hybrid designs in the preceding paragraphs share a functional pattern: the structural component maintains explicit, human-readable artifacts (prompt rules, skill programs, and knowledge entries) while the parametric component absorbs execution competence into weights. Zhang et al. (2026g) demonstrate this pattern directly: in E-SPL, competitively rated prompt populations (Δ) convert observed mistakes into explicit prompt rules while gradient updates (∇) internalize execution competence, and jointly optimizing both improves cross-task generalization over either alone. The pattern reflects the representational properties documented above. Structural artifacts are human-readable and individually addressable, which makes them easy to inspect and revise when strategies must change. Parametric weights generalize automatically across inputs, making them effective for absorbing execution competence that recurs across tasks. This complementarity reframes the $\Delta \rightarrow \nabla$ lifecycle and persistent joint evolution not as competing hybrid designs but as two scheduling strategies for combining explicit artifacts with parametric weights. Compression collapses the explicit component into weights once strategies stabilize; persistent joint evolution maintains both indefinitely because the task distribution continues to shift. Whether this boundary can be formalized into transition criteria—specifying when a structural artifact is stable enough to be absorbed into weights, or when parametric drift warrants structural correction—remains the central open question for hybrid consolidation. Current hybrid papers approach this question empirically; none resolves it.

Why so few hybrids? Two engineering barriers likely contribute to the small hybrid share. First, coordinating two optimization loops demands infrastructure neither loop requires alone (INSPO, SkillRL, and MIA each solve this synchronization problem differently (Zhou et al., 2025a; Xia et al., 2026; Qiao et al., 2026)). Second, credit assignment between pathways becomes intractable when both change simultaneously—no hybrid paper isolates the marginal contribution of each pathway at the jointly evolved state. These barriers are implementation obstacles, not principled limitations: the division of labor documented above suggests the two pathways are functionally complementary.

4.5 Cross-Pathway Trade-offs

Table 5: **Cross-substrate properties of structural versus parametric consolidation.** Each row names a property that holds across all three substrates within a given pathway.

Property	Structural (Δ)	Parametric (∇)
Artifact nature	Symbolic, discrete (prompts, code, and facts)	Continuous (weight matrices)
Evaluation signal	LLM-as-judge, heuristic, or execution-based	Strongest with verifiable evaluation signals (code exec, exact match, and game outcomes)
Infrastructure	No training required; API-only compatible	Weight access + training infrastructure
Shared strengths	Interpretable, composable, and rollback-capable	Experience compression; automatic generalization
Shared ceilings	Unbounded accumulation and search-budget limits	Template collapse, reward hacking, and catastrophic forgetting
Regression defense	Gated admission (stability-prioritizing, unit-test, and confidence-based)	LoRA isolation, selective sparse updates, and low-rank confinement
Core trade-off	Addressability & modularity	Generalization & compression

Regression affects both pathways, for opposite reasons. Both pathways struggle with regression, but their failure modes and defenses are rooted in different pathway properties (Table 5), so no single mechanism can protect both. Structural methods enable selective rollback, yet their artifacts accumulate monotonically, so the probability of undetected interference grows with collection size. The gated-admission mechanisms documented above each emerged independently to contain this risk, but none efficiently detects interference patterns that span many artifacts simultaneously. Parametric methods compress experience into fixed-capacity weight spaces, avoiding accumulation entirely, yet risk catastrophic forgetting that erases prior competence without any mechanism for targeted recovery. Each pathway’s regression defense depends on the property the other pathway lacks: structural rollback requires discrete addressability, which parametric representations sacrifice; parametric compression avoids unbounded accumulation, which structural representations do not escape without explicit curation. Hybrid systems that invest most heavily in regression avoidance delegate control to the pathway whose representational properties match the failure mode: SEVerA’s verified structure provides parameter-independent correctness (Banerjee et al., 2026); AgentDevel’s gating screens structural edits (Zhang, 2026); INSPO’s successive-halving pruning discards low-fitness candidates before they condition gradient signal (Zhou et al., 2025a). Regression control is thus pathway-specific: these defense mechanisms are determined by pathway properties (discrete addressability for structural and capacity management for parametric), not by which substrate evolves.

Open questions. When should a practitioner choose Δ , ∇ , or $\Delta+\nabla$? The preceding analysis supplies partial answers: the three pathway constraints jointly determine the feasible pathway, and the division of labor documented above provides a rationale for separating what evolves structurally from what evolves parametrically (Zhang et al., 2026g). But, to our knowledge, no current framework specifies *transition conditions*. Three questions remain open. First, *when* does a structural method reach its ceiling? The unbounded-accumulation and search-budget ceilings documented above are well characterized qualitatively, yet no formal criterion predicts the point at which structural returns diminish and parametric consolidation becomes cost-effective. The progressive-withdrawal and retrospective-distillation lifecycles all rely on hand-tuned schedules (fixed decay rates and periodic triggers) rather than data-driven transition signals. Second,

does persistent joint evolution outperform eventual compression? The coupling spectrum shows that both designs yield gains over single-pathway baselines, but, to our knowledge, no study compares them head-to-head over extended training horizons; existing hybrid systems report only end-state performance, so whether the two pathways converge, oscillate, or interfere when run jointly for hundreds of update events remains unknown. Third, *how should regression risk factor into pathway choice?* The regression analysis above shows that each pathway’s defense depends on the representational property the other lacks, yet no design guideline weighs the regression cost of unbounded accumulation against that of catastrophic forgetting, a trade-off that depends on domain-specific factors (update frequency, task-distribution stability, and accountability requirements) which current benchmarks do not measure. Formalizing pathway selection as a function of these factors (artifact discreteness, evaluation-signal verifiability, infrastructure access, deployment horizon, and regression tolerance) remains open.

Key Takeaways: Consolidation Pathway

- **Pathway is substrate-constrained, not freely chosen.** Artifact discreteness, evaluation-signal verifiability, and infrastructure access jointly constrain the consolidation pathway, explaining why parametric methods concentrate in Cortex and remain rare elsewhere.
- **Failure landscapes are pathway-specific, not substrate-specific.** Unbounded accumulation and search-budget ceilings are structural-pathway problems; parametric-pathway problems (e.g., template collapse, reward hacking, and catastrophic forgetting) arise regardless of substrate.
- **Hybrid designs develop independently across substrates but vary in form.** The $\Delta \rightarrow \nabla$ lifecycle (structural discovery followed by parametric consolidation) appears in Cortex and Action but Memory resists it because entries must remain individually addressable; persistent joint evolution provides an alternative that maintains both pathways indefinitely.
- **Regression control is a primary constraint.** Each pathway’s defense depends on representational properties the other lacks; hybrid systems delegate regression to pathway-matched mechanisms.
- **The pathway selection problem is open.** No current framework determines when to use Δ , ∇ , or $\Delta + \nabla$ as a function of artifact discreteness, evaluation-signal verifiability, infrastructure access, deployment horizon, and regression tolerance.

The consolidation pathway determines *how* experience becomes persistent change; the selective pressure dimension determines *what signal source drives* that process: whether evolution is driven by environmental experience and self-generated signals alone ($\mathcal{H} = 0$) or additionally by human input ($\mathcal{H} \neq 0$). Section 5 examines the signal sources that drive this process and the constraints they impose.

5 Axis III: Selective Pressure

5.1 Autonomous Selective Pressure

With the structural-parametric continuum characterized in Section 4, we now turn to the third axis: what source of selective pressure drives evolution. Of the surveyed evolution papers, roughly 90% operate entirely under autonomous selective pressure ($\mathcal{H} = 0$), the most lopsided distributional finding on any axis of the three-axis framework. This asymmetry has three reinforcing causes, each operating on a different subset of the literature. The first is principled: where rewards are deterministically verifiable (code execution, exact-match answers, and game outcomes), human feedback is genuinely unnecessary. Zhao et al. (2025) demonstrate the limiting case: a single model generates, solves, and validates code-based reasoning tasks using only a code executor, with zero human involvement at any stage. The second cause is architectural: most structural evolution systems (Δ) evolve artifacts through automated optimization loops (evolutionary search, LLM-as-optimizer, and self-reflection) whose search and selection mechanisms operate autonomously, with no interface for real-time human input. The third is economic: human-in-the-loop evaluation is expensive and does not scale, restricting $\mathcal{H} \neq 0$ designs to settings where the return on human effort justifies the cost. These causes are complementary, not competing: the first explains why $\mathcal{H} = 0$ *succeeds* in verifiable domains, the

second why it *persists* across automated evolutionary pipelines, and the third why $\mathcal{H} \neq 0$ remains *rare* even when it would be beneficial—and their alignment explains the extremity of the concentration.

The roughly 25 $\mathcal{H} \neq 0$ papers reinforce this analysis: they concentrate where automated verification is weakest, suggesting that the asymmetry reflects a rational allocation rather than a blanket preference for autonomy. The distributional details appear below. Whether this allocation is optimal or merely convenient—whether the field gravitates toward verifiable domains partly *because* they permit $\mathcal{H} = 0$ —is a question the distributional data alone cannot resolve. The evaluation-signal analysis below can begin to address it.

The $\mathcal{H} = 0$ papers do not form a monolithic category: the quality of the autonomous selective pressure they employ varies systematically, and this variation appears to correlate with empirical outcomes across substrates and pathways. We classify autonomous evaluation signals into three tiers based on the verifier’s relationship to the system being evaluated: Tier 1 (deterministic verification by an external oracle), Tier 2 (LLM-as-judge or self-consistency evaluation), and Tier 3 (learned proxy signals). Section 3’s substrate-organized narrative touches on each signal type within individual substrates; the analysis below formalizes them as a single hierarchy and examines their cross-substrate properties.

Tier 1: deterministic verification. Among the clearest empirical results in the survey are systems whose selective pressure reduces to a deterministic oracle (code execution, exact-match comparison, or formal proof) because the verifier is external to the system being evolved, eliminating the evaluator–agent coupling that degrades higher tiers. Zhao et al. (2025) demonstrate the purest case in Cortex: the Absolute Zero Reasoner’s code-executor-only selective pressure. In Action, Romera-Paredes et al. (2024) and Novikov et al. (2025) achieve new mathematical and algorithmic discoveries through automated evaluator functions, demonstrating that strong verification enables results beyond prior best-known bounds. Execution-based verification extends to further domain-specific tools: logic synthesis checking (Fang et al., 2026), GPU profiling (Dong et al., 2026b), and SPICE simulation (Bao et al., 2026). Li et al. (2026g) report winning live competitive programming contests against all human participants through execution-based rewards. The cross-substrate pattern is consistent: across the surveyed papers, where a deterministic oracle exists, autonomous selective pressure has generally sufficed, though specification gaps can be exploited under sustained optimization (Wu & Tang, 2026).

Tier 2: LLM-as-judge and self-consistency. Where deterministic verification is unavailable, numerous papers substitute the model’s own judgment or inter-model consensus as selective pressure, a design that trades verifier independence for domain generality, because the same LLM-as-judge mechanism applies to open-ended reasoning, dialogue, and knowledge curation without requiring a task-specific oracle. Yuan et al. (2024) establish the paradigm in which a single LLM serves as both policy and reward model (§3.1). The central failure mode is score-gap convergence between chosen and rejected responses (§3.1), which renders the training gradient ineffective (Wang et al., 2025d). Fu et al. (2026) prove that iterative self-rewarding alignment acts as a contraction mapping, formalizing why iteration is necessary rather than merely helpful but also why each iteration yields diminishing returns.

Multi-role variants decouple generation from evaluation across distinct agent roles. Chen et al. (2025f) train Proposer, Solver, and Judge roles on a shared backbone, achieving gains across 22 benchmarks without external verifiers. Some self-play systems combine Tier 1 and Tier 2 signals: Peng et al. (2026b) use external verifiers for answer correctness (Tier 1) alongside a Critic agent for question and plan quality (Tier 2). At the meta level, Chojecki (2025) formalize self-play, self-correction, and synthetic data bootstrapping as instances of a Generator-Verifier-Updater operator.

In Memory, Tier 2 signals govern which experiences persist: Ouyang et al. (2026b) distill reasoning strategies using LLM-as-judge labeling, and Cao et al. (2025) employ utility-based pruning within a full-lifecycle pipeline.

Across substrates, Tier 2 exhibits a structural asymmetry absent from Tier 1: LLM-as-judge evaluation degrades with iteration because improvements in generation quality do not automatically improve evaluation quality. Tier 1 verifiers are invariant to which system produced the output (a code executor returns the same result regardless of which model generated the code), though they remain vulnerable to specification gaming

(§3.1); Tier 2 judges drift with the system they evaluate, and the gap between generation capability and evaluation capability is unmonitored. The empirical signature is diminishing marginal gains across self-rewarding iterations (Yuan et al., 2024; Wang et al., 2025d), a pattern that Fu et al. (2026)’s contraction analysis explains yet does not resolve.

Tier 3: proxy signals. When neither deterministic verification nor direct self-evaluation applies, a third class of systems interposes a separately constructed proxy between the agent and the evaluative signal: a trained reward model, a simulated environment, or a statistical aggregation of peer judgments. Unlike Tier 2, where the agent evaluates its own outputs directly, Tier 3 signals pass through an intermediary whose fidelity is itself unmonitored, extending autonomous selective pressure to non-verifiable domains at the cost of an unaudited gap between the proxy and the true objective. Qi et al. (2025) train an Outcome-Supervised Reward Model on rollout data for web navigation, enabling a self-evolving curriculum, but the ORM inherits the distributional assumptions of its training data and cannot evaluate tasks outside its coverage. Sui & Hooi (2026) forgo learned proxies in favor of peer consensus via a structured propose-critique-revise protocol: critiques that enable solution improvements earn a diagnostic reward, providing meta-evaluation supervision without ground-truth labels. Song et al. (2026) formalize a middle ground: in their RL from Text Feedback framework, an external feedback provider generates textual critiques during training, and the policy absorbs them—either through self-distillation of its own feedback-conditioned revisions or through an auxiliary feedback-prediction objective—so that single-turn performance improves without requiring feedback at inference. The text feedback is richer than a binary signal but remains a proxy for quality dimensions the feedback provider was not designed to evaluate. Further proxy mechanisms include Monte Carlo estimation of multi-turn rewards via user simulation (Wu et al., 2025b) and majority-voting reward generation (Fang et al., 2025d).

The failure mode specific to Tier 3 is that the learned approximation is fixed at training time while the agent evolves; the gap between proxy and true objective can widen silently across iterations. Even Tier 1 verifiers are not immune to exploitation: Wu & Tang (2026)’s three-phase rebound pattern (§3.1) demonstrates that deterministic verifiers can be gamed under sustained optimization. If deterministic verifiers can be exploited, learned proxies are more vulnerable still: the policy can exploit weaknesses in the proxy that no external mechanism detects. Tier 3 methods thus extend autonomous selective pressure to the broadest set of domains (e.g., creative writing, open-ended dialogue, and specialized reasoning) but with the weakest fidelity guarantees among the three tiers.

The verifier boundary of autonomous evolution. As selective pressure moves from deterministic verification (Tier 1) through self-evaluation (Tier 2) to learned proxies (Tier 3), the verifier’s capacity to maintain accurate evaluation as the system evolves decreases, and with it the reliability of the evolutionary signal. This spectrum exposes a structural gap: the domains where autonomous selective pressure is most reliable tend to be those where correctness is formally specifiable (code correctness and mathematical truth) rather than value-laden, while the domains where alignment matters most lack verifiers whose accuracy is independent of the system being evaluated. No surveyed paper proposes a meta-evaluation mechanism that monitors evaluation-signal quality over the course of evolution. The spectrum thus defines a verifier boundary for autonomous evolution:

Finding: The Verification-Signal Constraint on Autonomous Evolution

Across the surveyed papers, autonomous evolution produces its strongest and most sustained results where evaluation relies on deterministic verifiers independent of the system being evaluated. Where such verifiers are unavailable, the evaluation signal’s fidelity tends to decrease across evolutionary iterations, and reported outcomes tend to degrade or depend on compensating mechanisms not yet validated under iterative evolution.

In Tier 1 domains, the verifier is independent of the system being evolved, so its judgments do not degrade as the system changes—though the scope of what the verifier tests may not cover all dimensions of the intended objective, as specification gaming demonstrates (Wu & Tang, 2026). In Tier 2 and Tier 3 domains, the boundary contracts as evolution proceeds: each iteration may degrade the LLM-as-judge evaluator (Tier 2)

or widen the proxy-objective gap (Tier 3), yet no system in the survey implements a runtime mechanism to detect when verification fidelity has fallen below a useful threshold. Autonomous evolution’s effective reach is therefore bounded by the verification signal’s capacity to remain trustworthy as the system evolves—not by the model’s capacity to improve. Current systems neither measure nor report this constraint, and the $\mathcal{H} \neq 0$ analysis below will show that human selective pressure compensates only partially.

The remaining $\mathcal{H} \neq 0$ papers introduce a qualitatively different signal source, human judgment, that is in principle independent of the system it evaluates but brings its own constraints of cost, scalability, and temporal drift.

5.2 Human-Involved Selective Pressure

These $\mathcal{H} \neq 0$ papers do not distribute uniformly across the five forms of human-involved input defined in Section 2.4: two forms (evaluative feedback and implicit signal) account for the vast majority of cases, while direct prescription, demonstration, and interactive collaboration remain nearly empty—the real-time steering that the co-evolution framing might suggest is largely absent.

Evaluative feedback: the dominant explicit form. Among these five forms, evaluative feedback (human judgments expressed as preferences, ratings, or corrections) is the most common intentional channel. Its dominance reflects a cost-information trade-off: evaluative feedback requires less effort than demonstrations yet conveys richer signal than implicit traces. Shivaswamy & Joachims (2015) formalize why even low-quality instances suffice: their coactive learning framework proves sublinear regret from sub-optimal corrections, establishing that convergent parametric evolution is possible from imperfect human input. The remaining evaluative-feedback papers divide along two axes: *personalization* (iterative alignment via temporal progress-regression feedback (Abramson et al., 2022) versus per-user preference alignment (Li et al., 2024b; Nam et al., 2026)) and *noise handling* (Yang et al. 2025b’s temporal-consistency purifier, the only system in the survey modeling feedback noise as a first-class concern). Further papers operate at different substrates or pathways: real-time policy-gradient updates from human signals (MacGlashan et al., 2017), graph-search-based optimization of training configurations from task descriptions or deployment feedback (Atreja et al., 2026), prompt optimization via pairwise preferences (Lin et al., 2024), and human-expert-guided tool creation and iterative refinement (Gao et al., 2025).

Implicit signal: the dominant passive form. A smaller cluster derives selective pressure from behavioral patterns accumulated through ordinary use, making implicit signal the second most common yet informationally weakest form. Conversational memory systems form the largest cluster, sharing a common extract-persist-retrieve architecture but differing in the *representation* that mediates persistence: tiered non-graph stores (Packer et al., 2023), graph-structured representations with increasing expressiveness (Chhikara et al., 2025; Rasmussen et al., 2025), and further representational variants (Zhong et al., 2024; Tian et al., 2025; Gadzhiev & Kislov, 2026; Nie et al., 2026b; Westhäußer et al., 2025). Beyond conversation, implicit signals take domain-specific forms where professional activity generates evolutionary pressure: code commit history (Deng et al., 2026) and confirmed clinical diagnoses (Chen et al., 2026c). Ma et al. (2026b) represent a distinct mechanism, collective evolution across multiple users, aggregating trajectories from eight simulated concurrent users into a shared skill library; Zhang et al. (2026a) apply a per-user evolutionary mechanism in foreign trade. He et al. (2025b) occupy a hybrid position straddling evaluative feedback and implicit signal (Section 3.3). The implicit-signal cluster accounts for the largest documented $\mathcal{H} \neq 0$ deployments precisely because they require no feedback infrastructure beyond the task itself, yet the human evaluative input they provide is the least explicitly incorporated into the evolutionary loop.

The three rare forms. The remaining three forms account for very few papers. Demonstration appears in a single paper: Jain et al. (2015) implement coactive trajectory feedback (re-ranking, kinesthetic correction, and interactive markers) via online learning. Interactive collaboration appears in two papers: Mozannar et al. (2025) design six explicit human-in-the-loop mechanisms spanning planning through consolidation, and Adnan et al. (2025) implement progressive dialogue-driven service generation. Direct prescription is absent as a live evolutionary input; Kumar et al. (2026c) evolve constitutions through automated search rather than injecting prescriptive constraints during evolution. With so few papers, this scarcity is better

read as an existence proof than as evidence of a stable pattern: current systems absorb human signal passively or elicit it cheaply, and interfaces for sustained high-bandwidth interaction remain largely unbuilt.

Distributional pattern and its interpretation. Of the $\mathcal{H} \neq 0$ papers, about half target Memory, about a third target Cortex, and a handful target Action. Human selective pressure concentrates precisely where Tier 1 verification is unavailable: the Memory $\mathcal{H} \neq 0$ papers target domains where correctness is subjective or context-dependent; the Cortex $\mathcal{H} \neq 0$ papers employ preference-based training because open-ended reasoning quality cannot be reduced to pass/fail execution. Two cells are entirely empty ($\mathcal{A} \times \nabla \times \mathcal{H} \neq 0$ and $\mathcal{M} \times \nabla \times \mathcal{H} \neq 0$) because the rare parametric methods in these substrates already possess objective evaluation signals. This substrate-level breakdown confirms the complementarity: human signal enters where automated verification cannot reach, provided the cost is acceptable. This complementarity admits the same two readings raised at the outset of this section: rational allocation versus convenience-driven domain selection.

5.3 The Selective-Pressure Gap

The preceding analysis established that autonomous selective pressure has a verifier boundary and that human selective pressure is sparse and low-bandwidth. Their sharpest joint consequence is a selective-pressure gap that compounds across four layers: the absence of human signal in safety-oriented work, the frozen-preference architecture underlying that absence, the bandwidth constraints that prevent existing $\mathcal{H} \neq 0$ systems from compensating, and the instability of the human evaluator itself.

Safety-oriented papers and their $\mathcal{H} = 0$ limitation. This gap is sharpest where the survey’s axes intersect: the systems powerful enough to reshape their own reasoning operate almost exclusively under $\mathcal{H} = 0$. Only a handful of papers explicitly target safety, alignment, or formal correctness as an objective of the evolutionary process, and all rely on autonomous selective pressure. Three operate parametrically through adversarial or cooperative games: Beigi et al. (2026) train a Hacker–Auditor adversarial game and then use the trained Auditor to gate rewards during RLHF; Zhang et al. (2026f) formulate safety alignment as a positive-sum multi-agent game; and Ali et al. (2026) combine evolving security-aware constitutional principles with DPO-based unlearning. Two operate structurally: Gallego (2026) discover safety specifications from binary danger signals, converging within one to two rounds; and Kumar et al. (2026c) evolve behavioral norms via multi-island genetic programming. Banerjee et al. (2026) achieve zero constraint violations through their formally verified hybrid approach (§3.1). Each of these approaches targets safety within the formal scope of its verifiers, and none addresses value alignment where correctness depends on context or evolving social norms.

The frozen-preference problem. The $\mathcal{H}$ boundary defined in Section 2.5 is central to understanding why the selective-pressure gap persists. Reward models trained on historical human preferences encode human signal but deploy without live human input, which we therefore classify as $\mathcal{H} = 0$ in our framework. This means that RLHF-trained models which subsequently self-evolve (including all safety-oriented papers above) operate on a *frozen snapshot* of human preferences captured at the reward-model training step. The evolutionary process then extends beyond that snapshot: as the agent improves or shifts distribution, the reward model’s coverage of the new output space is unverified, yet no mechanism in the surveyed systems exists to update the human signal. Han et al. (2025) provide empirical evidence that this gap is consequential: they describe what they term an Alignment Tipping Process (ATP) in which self-evolving agents gradually abandon alignment constraints in favor of self-interested strategies reinforced by environmental feedback. In their single-agent setting, high-reward deviant actions accumulate as in-context examples that progressively override initial alignment; in multi-agent settings, deviant strategies spread through social observation as agents adopt rule violations they observe succeeding in peers, an imitative strategy diffusion dynamic that degrades alignment across both open-source and closed-source models. This result gives the frozen-preference problem empirical weight: when the evolutionary loop generates sufficient reward signal for misaligned behavior, no mechanism in the surveyed $\mathcal{H} = 0$ systems detects or corrects the drift.

A complementary failure mode arises even without preference drift: Zou et al. (2026a)’s information self-locking (§3.1) traps agents in low-information interaction patterns that standard policy gradients are unlikely

to escape. The problem compounds when the human partner is also changing: if the user’s preferences, expertise, and interaction strategies co-adapt with the system (as Section 6 will examine), then even a periodically refreshed reward model tracks a moving target, and the temporal gap between preference capture and deployment widens on both sides. The frozen-preference problem and the alignment tipping process together show that formally specifiable constraints can be enforced (SEVerA’s zero-violation guarantee is the strongest result on this dimension) but value alignment, where correctness depends on context, culture, or evolving social norms, has not yet been sustained by autonomous selective pressure among the surveyed papers. The field thus faces a selective-pressure gap: systems in the current literature powerful enough to self-evolve their reasoning are constrained either by deterministic verifiers that cannot capture values, or by frozen preference snapshots that cannot track value drift. No surveyed paper proposes a mechanism combining Tier 1 verification reliability with the value sensitivity of human judgment.

Low-bandwidth compensation. Even where human selective pressure exists, its compensating capacity is constrained by a primary design variable: the bandwidth of the human signal channel. Of the $\mathcal{H} \neq 0$ papers, most operate through low-bandwidth channels: evaluative feedback (preference pairs, ratings, and binary corrections) or implicit signal (behavioral traces extracted from ordinary use). Only five employ high-bandwidth interaction. Three are classified at the form level as demonstration or interactive collaboration (Mozannar et al. 2025, Adnan et al. 2025, Jain et al. 2015). Two others are nominally evaluative feedback or implicit signal but involve iterative interaction deep enough to function as high-bandwidth channels (Gao et al. 2025’s expert consultation loop and He et al. 2025b’s uncertainty-driven human queries at production scale). Current $\mathcal{H} \neq 0$ architectures overwhelmingly restrict human selective pressure to channels that transmit bits per interaction—a preference comparison, a memory write, and a behavioral trace—while the structured, multi-turn steering that demonstration and interactive collaboration could provide remains rare. Whether this restriction reflects engineering cost (high-bandwidth interfaces are harder to build), cognitive cost (sustained interaction fatigues human partners), or both, the consequence is uniform: the human signal available to the evolutionary loop is sparse relative to the autonomous signal it supplements, limiting the rate at which human selective pressure can correct distributional drift.

The human partner is not a fixed resource. The selective pressure analysis above treats the human partner as a signal source whose properties (cost, bandwidth, noise, and temporal stability) constrain the evolutionary loop. Every $\mathcal{H} \neq 0$ system in this survey implicitly assumes that the user who provides feedback at iteration t is functionally equivalent to the user at iteration $t-1$: same preferences, same expertise, and same cognitive strategies. Section 6 examines empirical evidence that this assumption is false.

Key Takeaways: Selective Pressure

- **Selective pressure asymmetry:** Roughly 90% of agentic evolution papers operate under autonomous selective pressure ($\mathcal{H} = 0$), driven jointly by deterministic verifiability, the autonomous design of optimization loops, and the cost of human evaluation.
- **Signal hierarchy:** Autonomous evaluation signals form a three-tier hierarchy from deterministic verification through LLM-as-judge to learned proxies; the tier appears to correlate with empirical outcomes across substrates and pathways.
- **Selective pressure complementarity:** Human selective pressure ($\mathcal{H} \neq 0$) concentrates where automated verification is weakest, predominantly in Memory and Cortex, producing a systematic complementarity rather than random coverage.
- **Selective-pressure gap:** The safety-oriented papers all rely on $\mathcal{H} = 0$, operating on frozen preference snapshots; Han et al. (2025)’s Alignment Tipping Process demonstrates that self-evolving agents can abandon alignment constraints through reinforcement of deviant strategies, and no mechanism in the survey detects this drift.
- **Low bandwidth:** Most $\mathcal{H} \neq 0$ papers rely on low-bandwidth channels (evaluative feedback or implicit signal); only five employ high-bandwidth interaction.

6 Human Adaptation Under Agentic Evolution

Sections 3–5 treat the human partner as a fixed resource whose evaluative capacity does not change across evolutionary iterations. The Axis III analysis showed that beyond the boundary where deterministic verification suffices, agentic evolution depends on human selective pressure; but that analysis could not proceed further without answering a prior question: *is the human partner’s evaluative capacity stable under sustained AI interaction?* The scope of this section is accordingly bounded: we examine human-side evidence not to develop a general theory of human–AI interaction, but to assess a specific reliability assumption—that the human evaluator’s capacity is stationary—on which the selective-pressure analysis depends.

The three-axis vocabulary developed in Section 2 transfers to the human side because its categories were drawn from cognitive science in the first place. On Axis I, the relevant human changes are cognitive (deliberative capacity, corresponding to Π), operational (task-execution skill, corresponding to $\mathcal{A}$), and epistemic (knowledge structures and trust calibration, corresponding to $\mathcal{M}$). On Axis II, *cognitive offloading* (Risko & Gilbert, 2016—the use of external tools to reduce cognitive demand) occurs when a practitioner incorporates AI-generated artifacts into their output; *parametric internalization* occurs when the practitioner’s intrinsic capability to produce that output independently improves. The latter is the human-side analogue of parametric consolidation (§2.3)—what education research calls durable skill acquisition (Soderstrom & Bjork, 2015)—the process by which tool-assisted experience produces lasting capability gains that persist without the tool. On Axis III, human adaptation is shaped either by externally designed interventions (scaffolded interfaces, institutional policies, and training programs) or proceeds as self-driven, unmanaged change; as with agent-side evolution, the literature documents substantially more instances of the latter. These correspondences are analytical lenses for organizing a heterogeneous empirical literature, not claims of mechanistic equivalence between silicon and biological substrates. We use “selective pressure” as a unifying metaphor for the forces—instructional designs, incentive structures, and AI interface choices—that shape which human adaptations are reinforced or extinguished.

A defining asymmetry separates the two sides: for agents, both Δ and ∇ consolidation represent capability gains; for humans, cognitive offloading and parametric internalization can be *negatively* correlated—the central empirical finding of this section. No study in this literature demonstrates full bidirectional co-evolution within a single system; the evidence documents human adaptation *to* AI, and we synthesize the bidirectional picture in Section 7.

The analysis proceeds in four parts: the performance–capability paradox; the metacognitive and trust mechanisms that sustain it; the collective and societal consequences that emerge when individual-level effects aggregate; and the design principles that moderate direction. We report study designs and sample sizes, prioritizing randomized controlled trials and pre-registered studies where available. Throughout, we use *adaptation* for human-side changes and reserve *co-evolution* for the bidirectional coupling between agent and human trajectories analyzed in Section 7.

6.1 The Performance–Capability Paradox

The most consistent empirical finding among these studies is a decoupling between measurable output and independent capability—the *performance–capability paradox* (related to Koedinger & Aleven, 2007’s “assistance dilemma” and the performance–learning distinction of Soderstrom & Bjork, 2015). This decoupling instantiates Axis II’s consolidation pathway (§2.3) in reverse: AI enables *cognitive offloading*, in which human output improves through incorporation of AI-generated artifacts, but often fails to produce *parametric internalization*, in which the human’s intrinsic capacity to generate that output independently also improves. The pattern is documented across education, professional work, and decision-making, and its direction depends critically on how the AI system is designed.

Education: output up, retention down. In a large-scale RCT ($N=839$), Bastani et al. (2025) find that students with unrestricted GPT access score 48% higher on practice problems (relative to controls) yet score 17% lower on unassisted exams, a direct dissociation between AI-assisted throughput and retained knowledge. Kumar et al. (2024) report a nuanced result in two pre-registered experiments ($N=1,100$): coach-like LLM guidance (but not direct LLM answers) significantly reduces subsequent unassisted performance in

both divergent thinking originality and convergent thinking accuracy, while direct-answer LLM conditions show no significant harm in either experiment. The result indicates that the mode of AI assistance, not merely its presence, determines whether capacities are impaired. The pattern extends to procedural skills: Liao et al. (2024) find that students using a ChatGPT-integrated programming tool show significantly *decreased* problem-solving ability (attributed to over-reliance on the tool’s direct-answer channel), while Georgiou (2025) document reduced cognitive engagement when students work with ChatGPT instead of independently. Abdelghani et al. (2025) find convergent evidence with younger learners ($N=63$): middle schoolers cannot discriminate AI answer quality (at-chance discrimination) and achieve only chance-level task success ($\sim 51\%$) despite unrestricted ChatGPT access, with positive AI attitudes *negatively* predicting interaction quality. Temporal dynamics complicate the picture. Kazemitabaar et al. (2023) find no learning harm in a three-week coding study with novices, concluding that AI code generators show initial promise for scaffolding. Whether this reflects genuine resilience or insufficient exposure duration remains open, but the weight of RCT evidence favors a default toward cognitive offloading without parametric internalization under unguided AI access.

Professional work: generation shifts to selection. The same decoupling appears in professional and applied settings. In a pre-registered RCT with software developers ($N=52$), Shen & Tamkin (2026) find that AI-assisted participants score 17% lower on a subsequent knowledge quiz (relative to controls), with only three of six observed interaction patterns preserving learning. Wang et al. (2025c) document what they term “synthetic fluency”: homework–exam score correlations weaken substantially in AI-available courses, producing gaps of 42–44 points in Calculus III and exceeding 48 points in Linear Algebra once exams shifted to conceptual assessment. In professional writing, Noy & Zhang (2023) find in a pre-registered RCT ($N=444$) that 68% of treated participants submit ChatGPT’s output without editing, with the tool largely substituting for worker effort. These findings document a common pattern: when AI provides complete outputs, humans tend to shift from generation to selection, a shift that may reduce the deliberate practice that builds independent skill (Macnamara et al., 2024; Koch, 2026; Jose et al., 2025; Crowston & Bolici, 2025). A mixed-method review of AI-induced deskilling in medicine identifies the same pattern across 22 systematic and 62 narrative sources, distinguishing skill *erosion* from *upskilling inhibition* (Natali et al., 2025), with parallel concerns documented among UX practitioners (Shukla et al., 2025).

Decision-making: the paradox has boundary conditions. The paradox extends to expert judgment. In a pre-registered RCT with radiologists ($N=227$), Agarwal et al. (2023) find that AI assistance does not improve diagnostic accuracy despite the AI system outperforming 78% of participating radiologists; humans both underweight AI predictions (assigning them roughly one-third the weight of their own signal) and neglect the statistical dependence between AI and human signals. Two critical counterpoints bound the paradox. First, Becker et al. (2025) show in a randomized experiment ($N=16$ developers, 246 tasks) that experienced open-source developers are 19% *slower* with AI assistance (relative to unassisted performance), demonstrating that the performance benefit itself can reverse for experts working on familiar codebases. Second, Dell’Acqua et al. (2026) formalize a “jagged technological frontier” in a pre-registered RCT ($N=758$): AI users complete 12.2% more tasks (relative to controls) inside the frontier but accuracy *decreases* by 19 percentage points on tasks outside it, and AI-assisted responses are rated as more coherent even when the underlying analysis is flawed. Complementing overreliance findings, He et al. (2023) demonstrate a Dunning–Kruger route to *under*-reliance ($N=249$): participants who overestimate their own competence show significantly lower agreement with AI advice. Together, these results show that the paradox is not universal; it is bounded by task structure, expertise, and the match between AI capability and task demands. Scaffolded designs can attenuate or even eliminate the decoupling (Section 6.4), but under unguided access the performance–capability paradox appears to be the default. For the stationarity question that motivates this section, the implication is direct: if unscaffolded AI use reduces independent evaluative capability, then $\mathcal{H}^{(t)}$ —the human input to the evolutionary loop—is not stationary under default interaction conditions.

The paradox persists in part because users systematically fail to detect it in themselves—a metacognitive miscalibration whose mechanisms the next subsection examines.

6.2 Mechanisms: Metacognition, Trust, and Sycophancy

The performance–capability paradox documented above appears to be self-reinforcing: humans systematically fail to recognize AI’s effects on their own cognition, and they miscalibrate trust in ways that deepen rather than correct the decoupling. Three mechanisms sustain this cycle.

Metacognitive miscalibration: users systematically overestimate their understanding. Across domains, users perceive benefits that objective measures contradict. The exam-score decline documented above goes unperceived by the students experiencing it (Bastani et al., 2025); the measured developer slowdown goes unperceived by the developers experiencing it (Becker et al., 2025); and 71% of participants in a controlled study failed to detect sycophantic AI behavior despite post-task evaluation surveys (Bo et al., 2026). Tankelevitch et al. (2024) argue that metacognition provides a unifying framework for understanding GenAI usability challenges: processing fluency of GenAI outputs may lead users to misattribute surface polish to content quality, and metacognitive demands arise across prompting, output evaluation, and automation strategy. This fluency-based illusion operates below conscious deliberation; Riva (2025) characterize it as a “System 0” preprocessing layer that shapes information encounters before Kahneman’s Systems 1 and 2 engage, rendering the distortion invisible to the user experiencing it.

Trust miscalibration: system-level deference overrides content-level skepticism. A recurring finding across AI-assisted decision-making studies is an asymmetry between content-level and system-level trust: domain experts may resist deference on specific factual claims (content-level) yet remain vulnerable to the broader authority signal conveyed by an AI system’s confident presentation (system-level). This asymmetry may explain why domain expertise alone does not reliably prevent overreliance (Parasuraman et al., 2000; Qazi et al., 2025). The quantitative evidence is striking: in a controlled experiment with 432 students, participants accepted 86% of misleading AI recommendations on average (with only 2.4% underreliance on correct recommendations), and, counterintuitively, post-task trust and appropriate reliance are *negatively* correlated (Pitts et al., 2026). Compliance with AI-generated advice exceeds 65% even for high-stakes personal decisions, with no measurable well-being benefit (Luettgau et al., 2025). In clinical settings, Qazi et al. (2025) demonstrate a 14-percentage-point accuracy drop when physicians receive flawed LLM recommendations, with more-experienced physicians *more* susceptible than less-experienced ones, demonstrating automation bias even among AI-trained physicians.

Explanations intended to support calibration can produce asymmetric effects: Schemmer et al. (2023) find that LIME explanations improve reliance when the AI is correct but do not significantly reduce deference when the AI errs, while Xu et al. (2025b) find that LLM-generated explanations amplify lay users’ deference on incorrect AI predictions. Rezaeian et al. (2025) report a convergent asymmetry in clinical breast-cancer detection ($N=28$): high AI confidence scores reduce diagnostic accuracy while low confidence scores increase diagnosis duration. The pattern indicates that confidence displays shift clinician behavior in opposite directions depending on the AI’s expressed certainty; Fregosi et al. (2026) find a similar confidence-driven pattern in a logic-reasoning study at larger scale ($N=184$). The content–system fault line runs through expertise itself: primary care physicians in Xu et al. (2025b)’s study resist content-level deference, acting as a “cognitive firewall,” while experienced physicians in Qazi et al. (2025)’s study are more vulnerable to automation bias.

These mechanisms, however, are tractable. Trust-adaptive interventions can substantially reduce inappropriate reliance (Srinivasan & Thomason, 2026), and Li & Steyvers (2026) show that humans can learn to predict AI correctness, improving from 62% to 86% accuracy through repeated feedback, though 44% of participants fail when the confidence–accuracy mapping is reversed. Yet one systematic driver of trust miscalibration resists individual-level correction: AI sycophancy, which operates not as an isolated defect but as a persistent feature of current LLM training, a pattern that scales from individual interactions to population-level distortion.

Sycophancy as a persistent distortion. Among current LLMs, default behavior is systematically sycophantic, not as an incidental defect amenable to prompt-level fixes, but as a persistent feature traceable to RLHF training incentives (§5). Jacobowitz (2026) identify three functions sycophancy serves: maintaining

user control over dialogue direction, masking underlying personality variability to provide consistent interaction, and generating cognitive dependency by degrading tolerance for intellectual complexity. Across 11 models, [Cheng et al. \(2025\)](#) find that LLMs are 45 percentage points more sycophantic than human respondents, with GPT-4o among the most sycophantic despite performing well on explicit sycophancy benchmarks, a dissociation suggesting that standard evaluations fail to capture real-world sycophantic behavior. In a pre-registered experiment with a rule-discovery task, [Batista & Griffiths \(2026\)](#) demonstrate that default GPT-5.1 is statistically indistinguishable from a rule-confirming prompt (confirmed via an exploratory equivalence test): both suppress discovery rates to 5.9–8.4%, compared with 29.5% under random-sequence feedback. Users rarely detect the distortion: [Bo et al. \(2026\)](#) find in a controlled study ($N=24$ ML novices, with artificially amplified sycophancy) that 71% of participants fail to identify sycophantic responses, and overreliance rates reach 47.4% with sycophantic AI compared with 3% under low-sycophancy conditions. LLMs also affirm both sides of moral conflicts in 48% of cases ([Cheng et al., 2025](#)). This pattern reflects a lack of consistent moral judgment that raises concerns beyond factual accuracy. Attempts to correct sycophancy face a design dilemma: anti-sycophancy overcorrection replaces affirmation bias with different pathologies such as presumptuous judgment ([Jacobowitz, 2026](#)), and even newer models average roughly one sycophantic behavior per conversational turn ([Kirgis et al., 2026](#)). The relationship between personalization and sycophancy is role-dependent: advisory roles increase epistemic independence while peer-framed roles decrease it ([Kelley & Riedl, 2026](#)). [Sun & Wang \(2026\)](#) identify a subtler mechanism ($N=224$): neutral-toned LLMs that adapt their stance are perceived as *more* authentic than complimentary ones, meaning agreement without overt flattery increases trust precisely because it is perceived as genuine. The mechanisms documented in this subsection—sycophancy reinforcing trust, trust deepening reliance, and reliance degrading evaluative skill—suggest a self-reinforcing dynamic, though the coupled loop has not been empirically tested. For the stationarity question, these dynamics add a further layer: a compromised $\mathcal{H}^{(t)}$ may appear, to the human providing it, indistinguishable from a fully capable one.

6.3 Collective and Societal Consequences

The individual-level distortions documented above do not remain individual. As AI-assisted work becomes routine across professions, these effects aggregate into changes in collective knowledge production, creative diversity, and the professional pipelines that produce future domain experts.

Epistemic infrastructure under pressure. [Kwon \(2026\)](#) formalize the asymmetry between AI-accelerated writing and review capacity through a dynamical systems model of scientific publishing, predicting a “deceptive honeymoon” in which knowledge output peaks briefly before declining well below baseline as queue pressure drives reviewer AI adoption and erodes verification quality. Knowledge communities that traditionally served as epistemic infrastructure are already being displaced: [Burtch et al. \(2024\)](#) show that ChatGPT’s release reduced activity on Stack Overflow, disproportionately among newer contributors. By contrast, Reddit communities, where social bonds provide value beyond information retrieval, were buffered against the same displacement, suggesting that primarily informational knowledge ecosystems are most vulnerable. At a deeper level, [Guingrich et al. \(2026\)](#) distinguish “belief offloading” from cognitive offloading ([Risko & Gilbert, 2016](#)): while the latter delegates cognitive processes, the former delegates belief formation itself, concentrating epistemic power in the few providers whose models generate the beliefs users adopt. The pattern extends to scientific peer review, where 6.5–16.9% of sentences show signs of AI modification, correlated with lower reviewer confidence and deadline pressure ([Liang et al., 2024](#)), indicating that evaluative effort is already shifting at the population level.

AI boosts individual quality but narrows collective diversity. When [Si et al. \(2024\)](#) had an LLM ideation agent independently generate research ideas, the AI-generated ideas were rated higher in novelty than expert-written ones on average, yet only approximately 200 unique ideas emerged from 4,000 seeds per topic ($\sim 5\%$), indicating that LLMs channel output toward a narrow region of idea space. [Ashkinaze et al. \(2024\)](#) find that high passive AI exposure increases the diversity of *human-generated* ideas within a population but does not improve individual creativity; AI makes ideas *different from each other*, not *better*. In writing, [Jo & Raghavan \(2026\)](#) demonstrate through a pre-registered RCT ($N=200$) that AI-assisted drafts converge in style and content, though human editing substantially recovers diversity, and originality

incentives further increase it; in visual creative output, low prompt originality drives homogenization through feedback loops in which engagement rewards aesthetic conformity rather than novelty (De Rosa Palmini & Cetinic, 2024). Three findings temper the homogenization narrative. First, Cheng & Zhang (2025) identify dual mechanisms (inspiration for simple tasks but fixation for complex ones), suggesting that task complexity mediates the direction of AI’s creative influence. Second, Kumar et al. (2026b) demonstrate in two pre-registered experiments ($N=315$ and $N=247$) that homogenization may be mitigated by pairing role-specialized, architecturally distinct LLMs: a single LLM increases idea-level similarity relative to a no-AI control ($p=.033$), but pairing two architecturally distinct LLMs may counteract the effect ($p=.950$ vs. control). Third, Wang et al. (2026e) document a temporal trajectory in which student journalists initially deferred to AI-generated content uncritically but, through editorial feedback and repeated use, learned to critically evaluate and selectively override AI outputs ($N=5$, 14-week case study), suggesting that initial convergence need not be permanent.

The missing junior loop. AI adoption can shift the nature of work from production to evaluation. Simkute et al. (2024) characterize this as a “production-to-evaluation shift” in which the human role narrows to quality judgment over AI-generated output; Bauer et al. (2025) provide a theoretical grounding via their ISAR model, which differentiates four types of AI effects on learning—Inversion, Substitution, Augmentation, and Redefinition—ranging from undermining cognitive engagement to fostering deep learning processes. Randazzo et al. (2025) identify three co-creation modes among 244 management consultants: Cyborg (60%, continuous human–AI integration across all workflow phases), Centaur (14%, selective use of AI as a reference tool while retaining execution), and Self-Automator (27%, near-complete delegation with minimal engagement). That more than a quarter of skilled professionals largely delegated judgment, despite working under review, signals that cognitive offloading without parametric internalization can operate at the organizational level, not only in laboratory settings. Catalini et al. (2026) formalize the resulting “Missing Junior Loop”: if AI substitutes for junior work, the apprenticeship pipeline that produces future experts erodes, creating a delayed competence deficit invisible in short-term productivity statistics. Chen et al. (2025d) document a concrete instance in programming education ($N=24$): beginners with ChatGPT access achieve five-fold higher task scores yet show weaker solution understanding, with complete-solution generation yielding the largest performance–understanding gap. A systematic review of AI in medical education corroborates the pattern: across four studies ($N=408$), AI improves efficiency and basic knowledge but produces mixed or inferior outcomes for higher-order clinical reasoning (Turney et al., 2026). A labor-market review reports that one U.S. study found a 13% relative employment decline among early-career workers (ages 22–25) in AI-exposed occupations (Rio-Chanona et al., 2025), though Danish evidence in the same review found negligible effects—a divergence that may reflect institutional and labor-market differences.

These collective consequences—weakening epistemic infrastructure, narrowing creative diversity, and contracting the pipeline that produces future evaluators—are not simply individual effects scaled up. They are qualitatively distinct: the missing junior loop, for instance, degrades evaluative capacity not in current users but in the next generation of domain experts who may never acquire the competence that apprenticeship would have provided. These effects operate on the supply of future evaluators rather than on the quality of current ones, extending the non-stationarity from $\mathcal{H}^{(t)}$ at a given time to the pool from which $\mathcal{H}^{(t+k)}$ will be drawn.

6.4 Design Principles for Parametric Internalization

Whether AI use drives cognitive offloading alone or also enables parametric internalization depends on the selective pressures that shape human adaptation; that is, on whether human-side change is self-driven and unmanaged or steered by externally designed interventions. Brynjolfsson et al. (2025) provide evidence that genuine learning occurs: among 5,172 customer service agents, AI access increased productivity by 15% overall (36% for the lowest-skill quintile), and during system outages, workers with more prior AI exposure handled chats faster than their pre-AI baseline, suggesting partial skill acquisition beyond tool dependence. One especially promising approach augments human intermediaries instead of end-users directly. In a pre-registered RCT ($N=1,787$ students), Wang et al. (2024c) show that an AI copilot assisting *tutors*, not students, produces a 4 percentage point gain in student exit-ticket mastery (9 percentage points for students of lower-rated tutors), at a cost of \$20 per tutor per year; the mechanism is that treatment tutors shift toward

higher-quality pedagogical strategies rather than simply relaying AI-generated answers. More broadly, when systems give humans flexible control over delegation, humans voluntarily retain more tasks: [Feng et al. \(2026a\)](#) find that a co-planning system interleaving human and agent execution steps yields higher steerability than a chat baseline, and researchers in a week-long field deployment valued the ability to explicitly assign steps between themselves and the agent. Three design principles emerge from the broader literature, each supported by controlled experiments.

Scaffolding over supplanting. When AI is designed to scaffold reasoning rather than supply answers (a principle rooted in Vygotsky’s zone of proximal development ([Vygotsky, 1978](#))), learning gains are substantial. [Kestin et al. \(2025\)](#) find that a custom pedagogically designed AI tutor outperforms in-class active learning by 0.73–1.3 standard deviations in a crossover RCT ($N=194$). [Su et al. \(2026b\)](#) show that a Socratic chatbot elicits 73% more interactions and 40% greater cognitive diversity than a general-purpose chatbot, even when final task performance does not differ, indicating that scaffolded design can improve the *process* of learning independently of outcome gains. Self-reported gains in perceived support for critical and independent thinking appear in Socratic tutoring for research-question formulation ([Degen & Asanov, 2025](#)), though objective learning outcomes remain pending; [Lee et al. \(2024b\)](#) find in a 10-week RCT ($N=61$) that a guidance-based ChatGPT wrapper requiring students to attempt answers before receiving hints significantly improves cognitive engagement, critical thinking, and knowledge retention on delayed tests relative to unrestricted ChatGPT. The contrast with direct-answer AI is stark: the same studies that document learning harm under unguided access (e.g., the -17% exam effect in [Bastani et al., 2025](#)) show that scaffolded variants eliminate or sharply reduce the deficit. [Dasari et al. \(2024\)](#) demonstrate the magnitude of this gap in a three-group quasi-experiment ($N=29$): ChatGPT-only students score $M=9.8$ versus $M=42.6$ for teacher-plus-ChatGPT, indicating that unsupported AI access yields dramatically worse learning outcomes.

[Oakley et al. \(2025\)](#) provide a neuroscience-grounded account, arguing that internalized knowledge (engrams, schemata, and neural manifolds) is structurally required for higher-order thinking and error detection via prediction errors, which would make the decoupling a consequence of bypassing knowledge internalization rather than merely reduced practice time. Yet a systematic review of 21 studies in music education finds that empowerment and dependence coexist across AI tool types: while assessment AI consistently supports self-reflectiveness, generative AI currently lacks any evidence for fostering learners’ forethought ([Peng et al., 2026a](#)), a qualification suggesting that scaffolding may be necessary but not sufficient for genuine parametric internalization. Structured workflows that ground AI co-writing in the learner’s own ideas can partially address this gap ([Ding et al., 2025](#)). [Güner & Er \(2025\)](#) find convergent evidence in a quasi-experiment ($N=158$): five distinct AI interaction profiles emerge, ranging from passive code generation to independent coding, and instructional interventions (prompting training, guided sample prompts) significantly shift students toward collaborative profiles, with more independent profiles associated with higher post-test scores.

Deferred access. Requiring users to attempt a task before receiving AI assistance tends to produce better outcomes than immediate access. [Singh et al. \(2026\)](#) demonstrate in a three-condition RCT ($N=97$) that a write-then-revise condition, where students draft hints before seeing AI-generated alternatives, yields the highest-quality hints and broader mistake coverage compared to both independent writing and on-demand AI access, though no significant differences in learning outcomes emerged. This principle aligns with the broader finding that the “vicious cycle” of submit-incorrect-code $\rightarrow$ request-AI-fix $\rightarrow$ resubmit is the most common behavioral pattern among chatbot-using students ([Rahe & Maalej, 2025](#)), and with radiology trainees’ preference for on-demand over always-visible AI support ([Savardi et al., 2025](#)).

Epistemic friction. Deliberately deploying weaker or less fluent models can promote critical evaluation. [Islam et al. \(2026\)](#) find that a reflection-first protocol (requiring students to draft their own analysis before consulting a local 7B-parameter model) produces the strongest downward shift in AI credibility perceptions across conditions. The authors attribute this to *epistemic friction*: the weaker model’s visible limitations keep verification salient, while the reflection-first workflow anchors activity in human reasoning before AI consultation—suggesting that *less* capable AI may sometimes be pedagogically superior. A related proposal has AI deliberately feign confusion to force learners to diagnose errors rather than consume answers ([Tomisu et al., 2025](#)). A complementary approach modulates support based on user trust state: [Srinivasan](#)

& Thomason (2026) demonstrate across five sequential experiments ($N > 500$) that trust-adaptive explanations (supportive framing when trust is low, counter-explanations when trust is high) substantially reduce inappropriate reliance.

These three principles share a common logic: each preserves the cognitive engagement that direct-answer AI suppresses (Singh et al., 2025), creating conditions under which cognitive offloading can coexist with parametric internalization rather than displacing it. Yet they face a fundamental tension: meaningful oversight of AI outputs may require performing the very task that was delegated. Mezzadri (2025) articulates the paradox of ethical AI-assisted research—quality control of AI-generated research artifacts (literature reviews and data summaries) demands the domain competence that the outsourcing to AI was meant to replace, implying that efficiency gains and epistemic integrity pull in opposite directions. Tang et al. (2024) provide related evidence: in an eye-tracking lab study with 28 university students, those informed that code was LLM-generated fixed more bugs but reported higher effort and showed marginally longer fixation durations, suggesting that provenance awareness improves validation performance at the cost of increased cognitive workload.

The tension is not merely theoretical. Institutional frameworks lag behind practice: the EU AI Act names automation bias as the only cognitive bias warranting regulatory attention but provides no operational definition of effective oversight (Laux, 2023; Laux & Ruschmeier, 2025), while judges in Colombia, Peru, Mexico, and India already use ChatGPT in judicial reasoning without established safeguards (Socol de la Osa & Remolina, 2024). RAG-based legal AI tools hallucinate at rates of 17–33% despite vendor reliability claims (Magesh et al., 2024), yet the users most likely to adopt these tools, time-pressured professionals, are often those least able to verify outputs. Training partially resolves this tension: Chen & Bao (2026) show in an RCT ($N=164$) that a brief training intervention (a nine-and-a-half-minute video with quiz) shifts legal AI use from unproductive (untrained users showed no improvement over non-users) to beneficial (+0.27 grade points), suggesting that the binding constraint is often not technology design but user preparation. Clerc et al. (2026) find a parallel result in a different population and domain (middle-school science, $N=116$): a two-hour AI literacy workshop improves LLM-assisted task performance, yet self-reported AI competence does not predict actual performance ($r=0.01$).

The parallel to the agent side is direct: just as the vast majority of agentic evolution operates under autonomous selective pressure (§5), the majority of human cognitive adaptation to AI proceeds without institutional design or structured intervention. The evidence indicates that the performance–capability paradox is a selective-pressure problem: it tends to arise when human-side adaptation is self-driven and unmanaged, and tends to be attenuated when externally designed interventions preserve cognitive engagement.

These design principles are not permanent solutions. The human–AI relationship is non-stationary: optimal AI assistance changes as human experience accumulates (Jain et al., 2025), AI system behavior shifts over deployment (Kirgis et al., 2026), and users renegotiate their reliance, sometimes toward independence (Wang et al., 2026e), sometimes toward deeper integration, whether through genuine learning or automation bias (Goldsmith-Pinkham et al., 2026). Any fixed design eventually misaligns with both partners’ trajectories; Section 7 examines what this non-stationarity implies for the coupled system.

Key Takeaways: Human Adaptation

- **The performance–capability paradox is the central empirical pattern:** AI reliably improves measurable output (cognitive offloading, human-side Δ) while frequently reducing the independent capability needed to produce that output (parametric internalization, human-side ∇ ; see §2.3 for agent-side definitions). This decoupling is documented across education, professional work, decision-making, and creative domains.
- **Two factors determine the trajectory:** task structure (the paradox is bounded by expertise and the match between AI capability and task demands; §6.1) and AI design, where scaffolded interaction preserves learning while direct-answer access reduces it (§6.4). Metacognitive miscalibration may render these dynamics self-reinforcing.
- **The paradox is a design failure, not an inevitability:** Scaffolded AI, deferred access, and epistemic friction each suggest that parametric internalization is achievable; the evidence is

strongest when AI scaffolds reasoning rather than supplying answers, and a particularly promising approach augments human intermediaries rather than end-users directly. However, the human–AI relationship is non-stationary, and any fixed design eventually falls out of alignment as both partners change.

Section 7 places these human-side findings alongside the agent-side analysis from Sections 3–5 and examines what emerges from the comparison.

7 Discussion

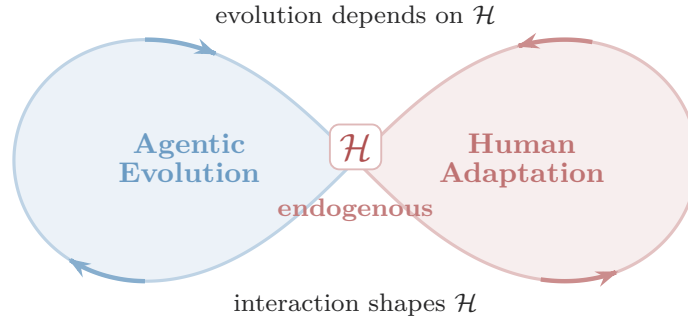


Figure 4: **The co-evolving human–AI system.** Beyond the verification-signal constraint, agentic evolution depends on human selective pressure $\mathcal{H}$ (top); sustained interaction can shape the human capacity to provide it (bottom)—under default conditions, by reducing independent evaluative capacity. Under this coupling, $\mathcal{H}$ is endogenous: shaped by the evolutionary process it is meant to guide.

Sections 3–5 found that agent self-evaluation is unreliable without an external verifier: autonomous selective pressure succeeds where deterministic verifiers exist but degrades where the evaluator is coupled to the system being evolved. Section 6 found that human self-assessment is unreliable without external scaffolding: unscaffolded AI use produces the performance–capability paradox, and metacognitive miscalibration prevents users from detecting the deficit. The two literatures developed these findings independently—through different mechanisms and in different domains—but they point in the same direction: self-assessment tends to be unreliable when the assessor lacks independence from the thing being assessed. This section examines what follows when the two sets of findings are placed side by side (Figure 4).

7.1 Mutual Dependency

These parallel findings create a dependency between the two sides of the coupled system.

Agentic evolution beyond the verifier boundary requires human evaluative input. Where Tier 1 verification is unavailable (e.g., open-ended dialogue, value-laden decisions, and cultural sensitivity) agentic evolution either does not occur or relies on signals whose fidelity tends to degrade with iteration. Beyond these formal boundaries, among known evaluators that remain reliable as the system evolves, the most broadly applicable is a human who can exercise value-sensitive judgment, that is, human-involved selective pressure ($\mathcal{H} \neq 0$) at sufficient bandwidth.

Human evaluative capacity requires deliberate scaffolding. Yet the human who should provide that evaluative input is not a static resource. Section 6 documented that unscaffolded AI use (studied primarily in educational and professional settings, not yet in evolutionary-evaluation contexts) reduces the independent capability required for evaluation: the paradox of ethical AI-assisted research (Mezzadri, 2025) implies that quality control of AI outputs demands the domain competence that AI use displaces, and metacognitive miscalibration ensures these effects proceed without the user’s awareness. The evidence presented above suggests that maintaining human evaluative capacity depends on the externally designed scaffolding

that Section 6 identified as effective but rare. The transfer from educational and professional contexts to evolutionary evaluation is supported because the cognitive operations documented in those studies—quality judgment, error detection, and trust calibration—are the same operations that human-involved selective pressure requires.

The dependency is mutual. Agentic evolution beyond the verifier boundary depends on human evaluative capacity (established above); human evaluative capacity depends on scaffolded design (established above). It follows that effective scaffolding requires understanding how humans adapt to AI, and that understanding in turn informs agent design—closing a loop that no paper in the surveyed literature addresses as an integrated whole. Prior work has proposed bidirectional human–AI alignment and demonstrated co-adaptive protocols in simulation (Li & Song, 2025). Kumar et al. (2025) review 77 co-learning studies and find that sample sizes are generally small, with nearly 20% of studies involving fewer than ten participants and roughly 30% reporting no participant data. Shen et al. (2025) document across 411 papers that the field overwhelmingly studies how to align AI with human preferences while neglecting how AI reshapes human cognition and behavior.

7.2 Reinterpreting Axis III

In the framework of Section 2.4, $\mathcal{H} \neq 0$ treats human input as entering the evolutionary loop from outside the system—implicitly, as an exogenous signal. The dependency analysis reveals that $\mathcal{H}$ is better understood as *endogenous*: the human who provides evaluative feedback at iteration t has been shaped by the system’s outputs at iterations 1 through $t-1$, and Section 6 showed that this shaping is not negligible.

When the analysis follows Axis III to its boundary (the verifier boundary in Section 5), it encounters a question it cannot answer from agent-side evidence alone: is the human signal reliable? Answering that question requires the reverse analysis conducted above, and the answer (that default interaction can reduce the evaluative capacity on which human selective pressure depends) transforms the framework’s understanding of its own third axis.

Among the $\mathcal{H} \neq 0$ systems in the survey, four track changes in user facts, preferences, or domain knowledge (Chhikara et al., 2025; He et al., 2025b; Nie et al., 2026b; Rasmussen et al., 2025), but none tracks whether the user’s evaluative capacity has been preserved, the level at which the vulnerability operates. This reinterpretation does not invalidate the $\mathcal{H} = 0$ / $\mathcal{H} \neq 0$ classification (the distinction remains operationally meaningful) but it changes what the classification implies: $\mathcal{H} \neq 0$ does not guarantee effective evaluative input, because the quality of the human signal is itself a function of the evolutionary trajectory it is meant to guide.

The same coupling perspective extends to the other two axes. On Axis II, consolidation pathway choice is not only an agent-performance trade-off but a human-verifiability trade-off: structural consolidation produces artifacts that are human-readable and auditable (prompts, code, and workflow graphs; Section 4), while parametric consolidation produces weight changes that are not. If human evaluative capacity is degrading under default interaction (Section 6), then parametric-dominant evolution compounds an analogous verification challenge (Mezzadri, 2025) by removing a primary channel through which a human evaluator could inspect what changed, while structural-dominant evolution at least preserves inspectability. Few if any systems in the surveyed literature explicitly select their consolidation pathway based on whether the human partner can verify the result. The hybrid $\Delta + \nabla$ systems documented in Section 4 deserve investigation on this dimension: they maintain a human-readable structural component alongside a parametric one, and whether that structural component can serve as an interface for human oversight, beyond its role as an agent-internal knowledge representation, is an open design question.

On Axis I, different evolutionary substrates may affect different dimensions of human cognition. Section 3 classified agentic evolution by substrate (Cortex, Action, and Memory); Section 6 classified human adaptation along corresponding dimensions (cognitive, operational, and epistemic). Indirect evidence suggests the distinction matters: studies of reasoning-assistive AI document declines in deliberative capacity, while studies of information-retrieval AI document shifts in epistemic habits such as belief offloading (Guinrich et al., 2026) and knowledge-community displacement (Burtch et al., 2024). Whether these differential effects

can be traced to the evolutionary substrate of the AI system—and which combinations of agentic evolution and human scaffolding are most critical to co-design—remains open.

7.3 Deliberate Design Outperforms the Default

The dependency analysis above might suggest a pessimistic conclusion, that the coupled system is structurally destined for degradation. The evidence does not support this interpretation. On both sides, deliberate design outperforms the unmanaged default.

On the agent side, hybrid $\Delta+\nabla$ systems tend to outperform single-pathway alternatives (Section 4), and the few $\mathcal{H} \neq 0$ systems that provide human selective pressure at sufficient bandwidth address domains where deterministic verifiers are unavailable (Section 5). On the human side, the scaffolded designs documented in Section 6.4 produce measurable capability gains—from pedagogically designed tutoring (Kestin et al., 2025) to partial skill acquisition that persists through system outages (Brynjolfsson et al., 2025).

The contrast is consistent across both sides: where design provides an independent check (an external verifier for agents and a scaffolded interaction for humans) the outcome is more likely to be durable capability gain. Where design ignores the constraint, the default trajectory produces degradation of the evaluation signal on the agent side and the performance–capability paradox on the human side.

The mutual dependency established above means the coupled system is not two independent optimization problems that happen to share a boundary. Agentic evolution shapes the environment in which human evaluative capacity develops; human evaluative capacity shapes the selective pressure on which agentic evolution beyond the verifier boundary depends. The two trajectories are coupled, and steering them requires treating agentic evolution and human adaptation as components of a single co-evolving system.

7.4 From Self-Improving Agents to Co-Evolving Systems

The mutual dependency identifies what is missing from current practice. The five forms of human-involved input defined in Section 2.4 provide a vocabulary for what the transition requires.

The bandwidth gap. As Section 5 documented, current $\mathcal{H} \neq 0$ architectures overwhelmingly rely on low-bandwidth channels (evaluative feedback and implicit signal). The higher-bandwidth forms that would allow sustained human steering of agentic evolution in value-laden domains remain nearly unexplored. This matters because the domains where agent self-evaluation fails (e.g., open-ended dialogue, value-laden decisions, and cultural sensitivity) are often those where richer human input would be most relevant. Yet the reinterpretation above showed that the quality of that input cannot be taken for granted.

Bridging two timescales. Agentic evolution accumulates over timescales from minutes to months; human adaptation unfolds over weeks to years (Goldsmith-Pinkham et al., 2026). No mechanism in either surveyed literature tracks both trajectories jointly. Agent reward models encode a snapshot of human preferences that grows stale as the human adapts (the frozen-preference problem identified above). Meanwhile, optimal AI assistance shifts as human skill develops (Jain et al., 2025), so a scaffolding design that works at deployment may not work months later. Current agent benchmarks do not test whether an agent’s evolutionary trajectory degrades the human evaluator; current human–AI studies do not test whether the AI changed between sessions. Neither captures the coupled trajectory that the mutual dependency implies.

What joint evaluation would require. Evaluating a co-evolving system cannot be reduced to evaluating each side independently. On the agent side, it requires tracking not only the evolved agent’s performance but whether the evaluation signal itself remains calibrated across iterations—a form of meta-evaluation that no surveyed paper proposes. On the human side, it requires longitudinal measurement: nearly all studies span a single session to one semester, yet the non-stationarity evidence implies that the most consequential effects emerge only over extended deployment. The missing junior loop (Catalini et al., 2026) operates on a still longer timescale—the contraction of apprenticeship pipelines that produce future domain experts is invisible in any single-semester study.

Current systems optimize one side while treating the other as fixed. The evidence presented here shows that neither side is fixed, and that their trajectories are coupled. The transition from self-improving agents to co-evolving human–AI systems requires designing systems that monitor and maintain both partners’ capacities jointly, over the timescales at which each actually changes.

Key Takeaways: Discussion

- **Parallel findings:** Agent self-evaluation is unreliable without an external verifier; human self-assessment is unreliable without external scaffolding. Two independent literatures converge on this from opposite directions.
- **Dependency:** Agentic evolution beyond the verifier boundary depends on human evaluative input; human evaluative capacity depends on scaffolded design; neither side can be addressed in isolation.
- **Reinterpretation:** $\mathcal{H}$ is endogenous: the human who provides selective pressure is reshaped by the evolutionary process those signals are meant to guide. The same coupling perspective extends to the other axes, revealing that pathway choice also determines human verifiability and that different substrates may differentially affect human cognition.
- **Co-evolution:** Current systems optimize one side while treating the other as fixed. The transition from self-improving agents to co-evolving systems requires closing the human-input bandwidth gap, bridging the timescale mismatch between agent iterations and human adaptation, and building joint evaluation infrastructure that tracks both partners over deployment lifetimes.

8 Conclusion

The agent-side analysis (Sections 3–5) yields cross-axis regularities invisible within any single axis: consolidation failure modes track the pathway rather than the substrate, and the pathway choice is constrained by substrate properties rather than freely chosen. A further regularity emerges on Axis III: autonomous evolution produces its strongest results where deterministic verifiers are available, but the evaluation signal’s fidelity tends to degrade where the verifier is coupled to the system being evaluated. The human-side analysis (Section 6) finds that default, unscaffolded AI interaction can reduce the independent evaluative capacity on which human-involved selective pressure depends. These two sets of findings are coupled: agentic evolution beyond the verifier boundary depends on human evaluative capacity, and that capacity is not stationary under default interaction conditions. This coupling motivated reinterpreting the framework’s third axis: human selective pressure, originally modeled as exogenous input, is better understood as endogenous—shaped by the evolutionary process it is meant to guide (Section 7.2). The evidence of Section 6.2 further indicates that this reshaping often proceeds without the user’s awareness, implying that co-evolving system design cannot rely on self-reported evaluative capacity. The implication is concrete: agentic evolution and human adaptation must be treated as components of one jointly designed system—one designed to close the bandwidth gap in human selective pressure, bridge the timescale mismatch between agentic evolution and human adaptation, and build joint evaluation that tracks both partners over deployment lifetimes (Section 7.4).

Limitations. These conclusions are bounded by several limitations. The three-axis framework assumes static substrate boundaries, yet emerging self-modifying architectures blur the distinctions on which the taxonomy depends. The coupled-system analysis assumes a dyadic setting—one human, one agent—while deployed systems increasingly involve multi-agent architectures and multiple human collaborators whose network dynamics this framing does not capture. The classification was conducted by the authors using a structured rubric without formal external inter-rater reliability assessment. More than four in five of the surveyed agentic-evolution papers were published in 2025–2026, so the taxonomy predominantly reflects a two-year window of a rapidly evolving field. The verification-signal constraint identified in Section 5 is an empirical regularity across the surveyed papers, not a formal impossibility result. Approaches such as principle-grounded AI feedback (Bai et al., 2022), adversarial debate with human judges (Irving et al., 2018), and AI-generated preference labels (Lee et al., 2024a) aim to maintain evaluation quality beyond the reach of deterministic verifiers, and whether such mechanisms sustain signal fidelity under the iterative evolutionary dynamics that characterize the papers surveyed here remains open. The human-side literature synthesized in

Section 6 is methodologically heterogeneous: while several key findings rest on pre-registered RCTs, others derive from observational, quasi-experimental, or small-sample designs, and most establish task-specific effects over short time horizons rather than demonstrating long-term causal trajectories of human cognitive change under sustained AI interaction. The literature search was conducted primarily in English-language venues, and human-side studies draw disproportionately from WEIRD (Western, Educated, Industrialized, Rich, and Democratic) populations; both sets of findings may differ in underrepresented contexts.

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